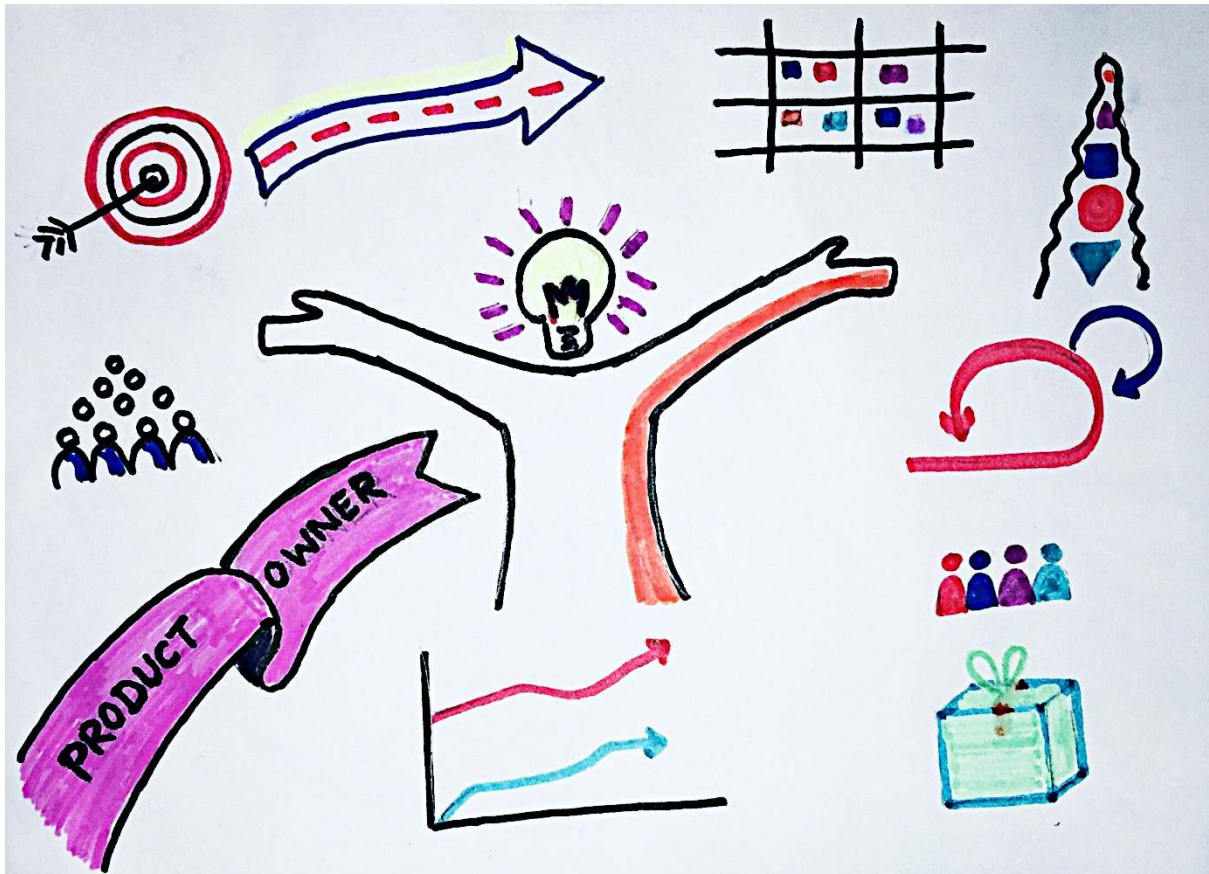


BUILD GREAT PRODUCTS That Always **Deliver** **VALUE** TO CUSTOMERS



Version 7.12

Based on Scrum Guide 2020 Version

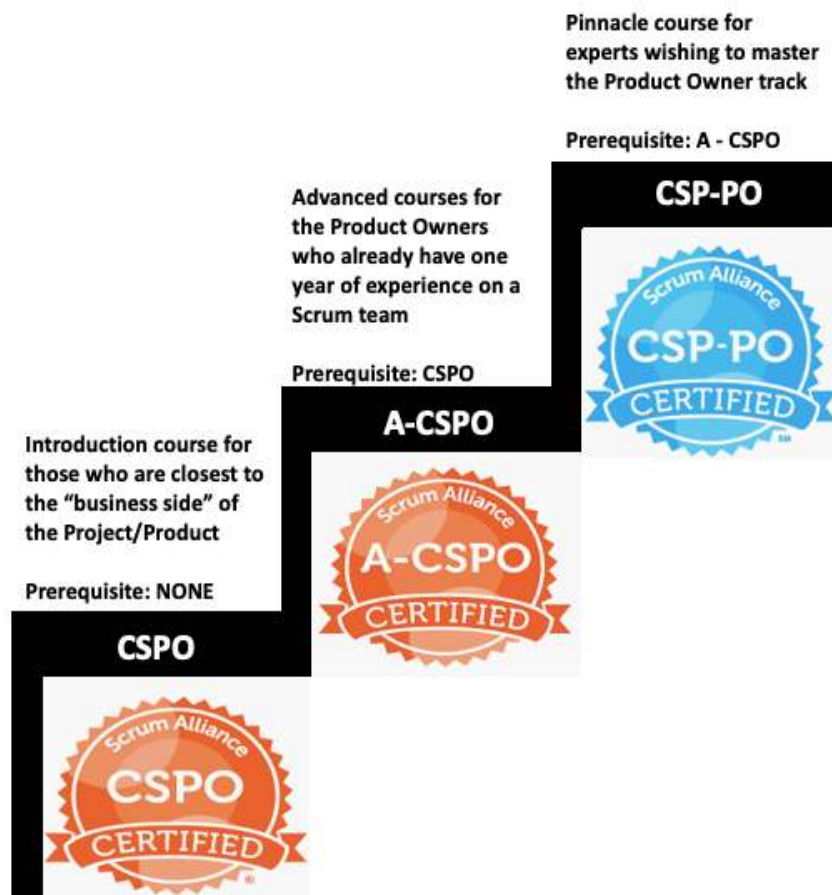
Trainer

VIJAY BANDARU

AGENDA

DAY 1 TOPICS	DAY 2 TOPICS
Forming Teams, Intro & Workshop setup	Product Discovery
Product and Product Levels	Agile Product planning
Product Lifecycle	Product Vision
Minimum Viable Product	Prioritization Techniques
Project & Product Mindset	Product Roadmap
Agile Manifesto Values & Principles	Product Backlog
Scrum Framework overview	Product Backlog Refinement
Scrum Framework Elements	User stories
Scrum Values and Roles	Release planning and Tracking
Scrum Framework Flow	Product increment
Sprint, Scrum Artifacts & Definition of Done	Release Planning and Tracking Simulation
Product Owner Role Deep Dive	Certification Discussion
Day 1 Wrap up	Closure

Scrum Alliance CSPO Certification Levels



Meet Your Trainer



VIJAY BANDARU

MY LEARNING JOURNEY ACCREDITATIONS





MY COMMUNITY INVOLVEMENT & RECOGNITIONS

MY VISION

Make Learning SPECIAL

S – Simple
P – Practical
E – Effective
C – Collaborative
I – Innovative
A – Adaptive
L – Long-lasting

- ✓ International Trainer
- ✓ Delivered 2 Products as Product Owner
- ✓ Training from the Back of the Room (TBR) certified
- ✓ Certified Agile Coach
- ✓ Experience in Large Scale Agile Transformation
- ✓ Received "Best Trainer" multiple times
- ✓ Volunteer, Speaker, and Organizer
- ✓ Published Several Articles
- ✓ Created "WONDER" Coaching Model

➤ My previous Trainings Videos, Photos and Feedback at:

www.facebook.com/learnovative

 www.vijaybandaru.com

 www.learnovative.com

 www.linkedin.com/in/vijaybandaru

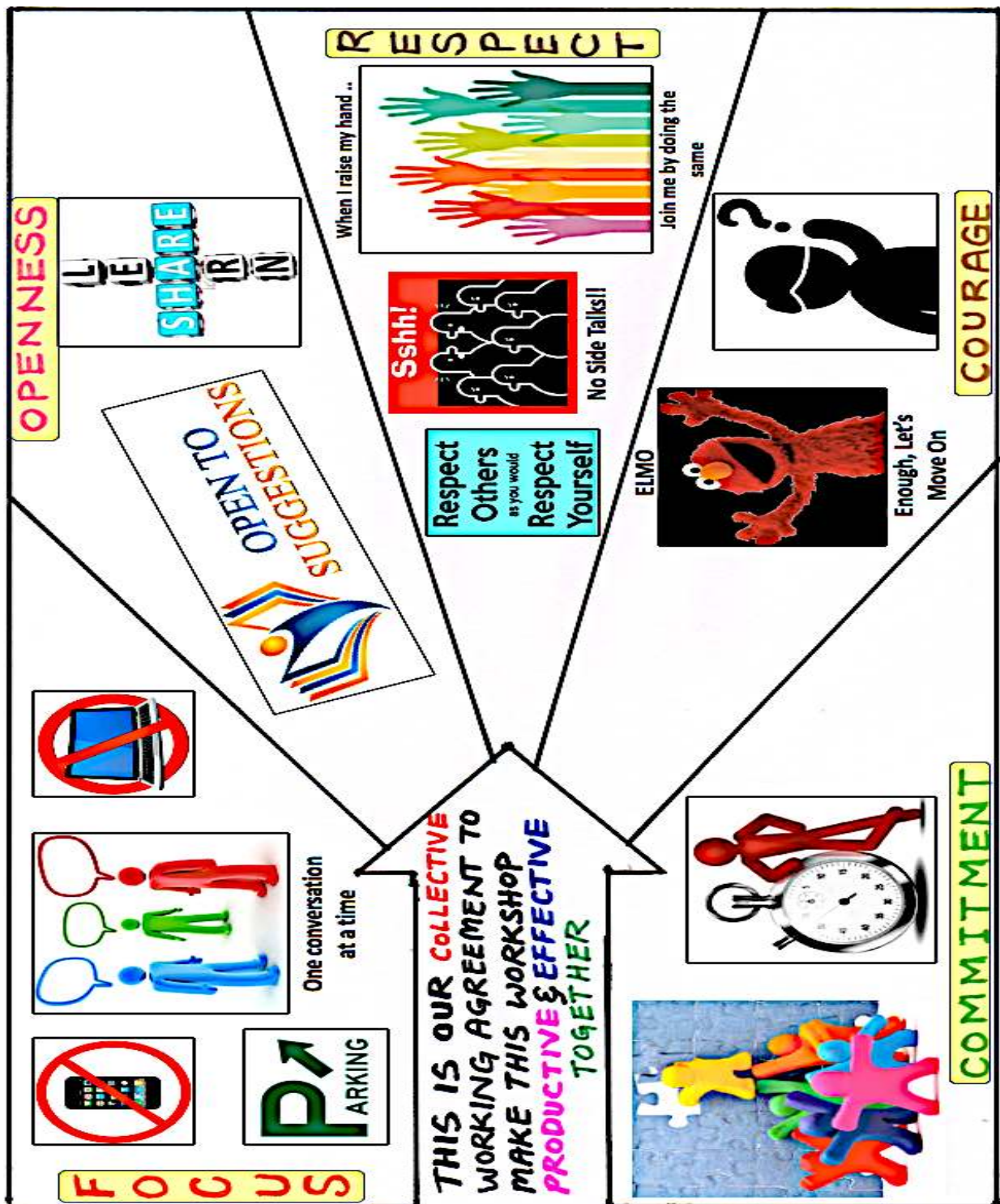
 vijaybandaru@learnovative.com

  **+91-98480 32144**

Training is not just my **PROFESSION**, it is my **PASSION**

I hope you enjoy the learning in my class and wish you good luck !!

Our working agreement



Quick overview:

1. Products should be **customer** centric
2. Products should be built to solve a **problem** or should fulfil a **need**
3. Products must be useful and also **usable**
4. Regular customer involvement during Product Development helps to receive early **feedback**
5. Early feedback helps you to keep cost of **change** under control
6. Product development should leverage the latest **technologies**
7. Products should have a clear **vision** and a strong **strategy**
8. Product owner should be **empowered** and **accountable** to own the end to end Product development and to make decisions faster

Focus on the "NEED" not the "REQUIREMENT"



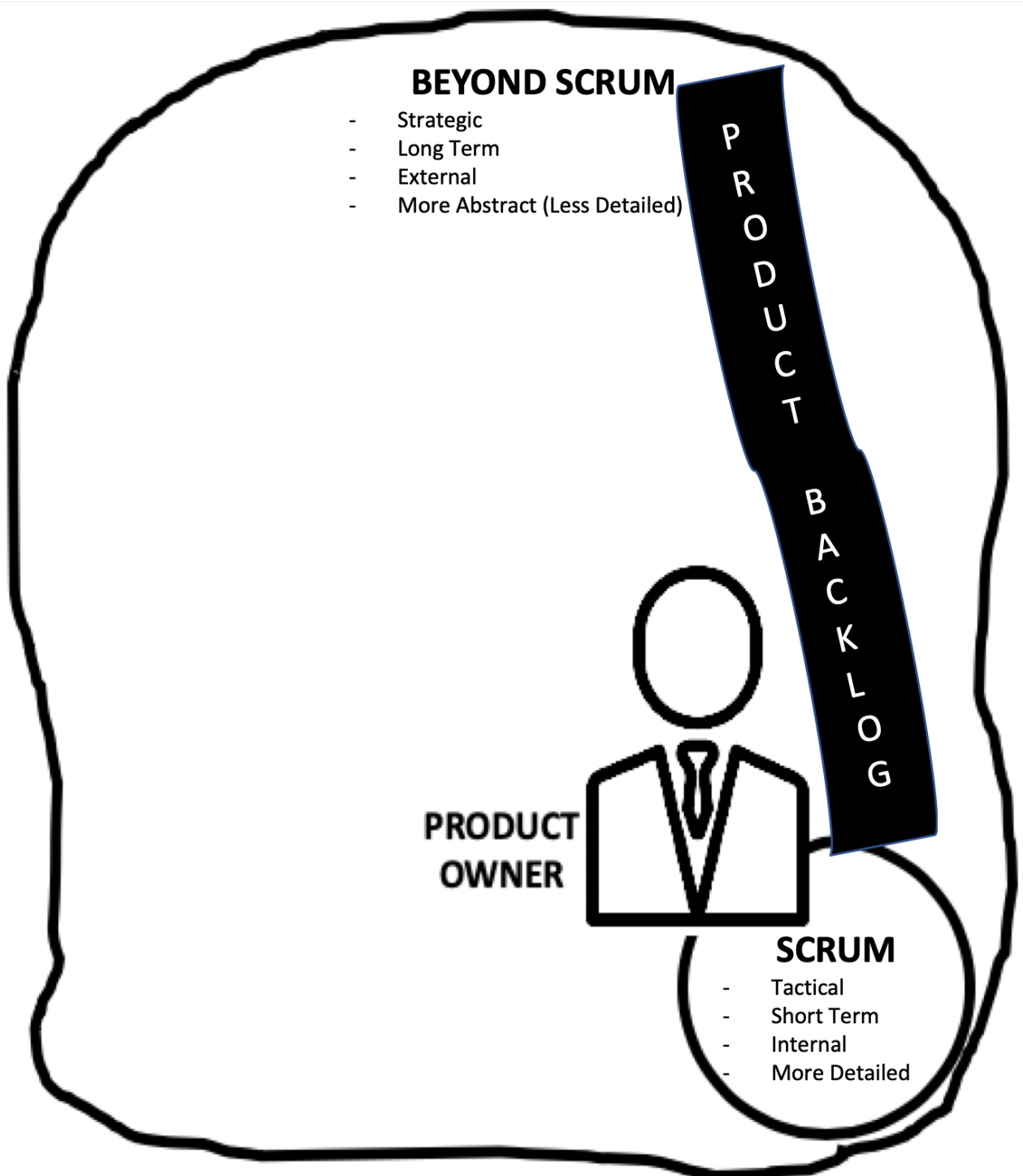
The need may be the customer has an 80 years old mother, she has been done with knee replacement surgery recently and she is unable to walk so she needs comfort. But the customer is rather than focusing on the need, she is focusing on requirement and the Product Owner is getting a wrong understanding.

This cup may be useful but it is NOT usable



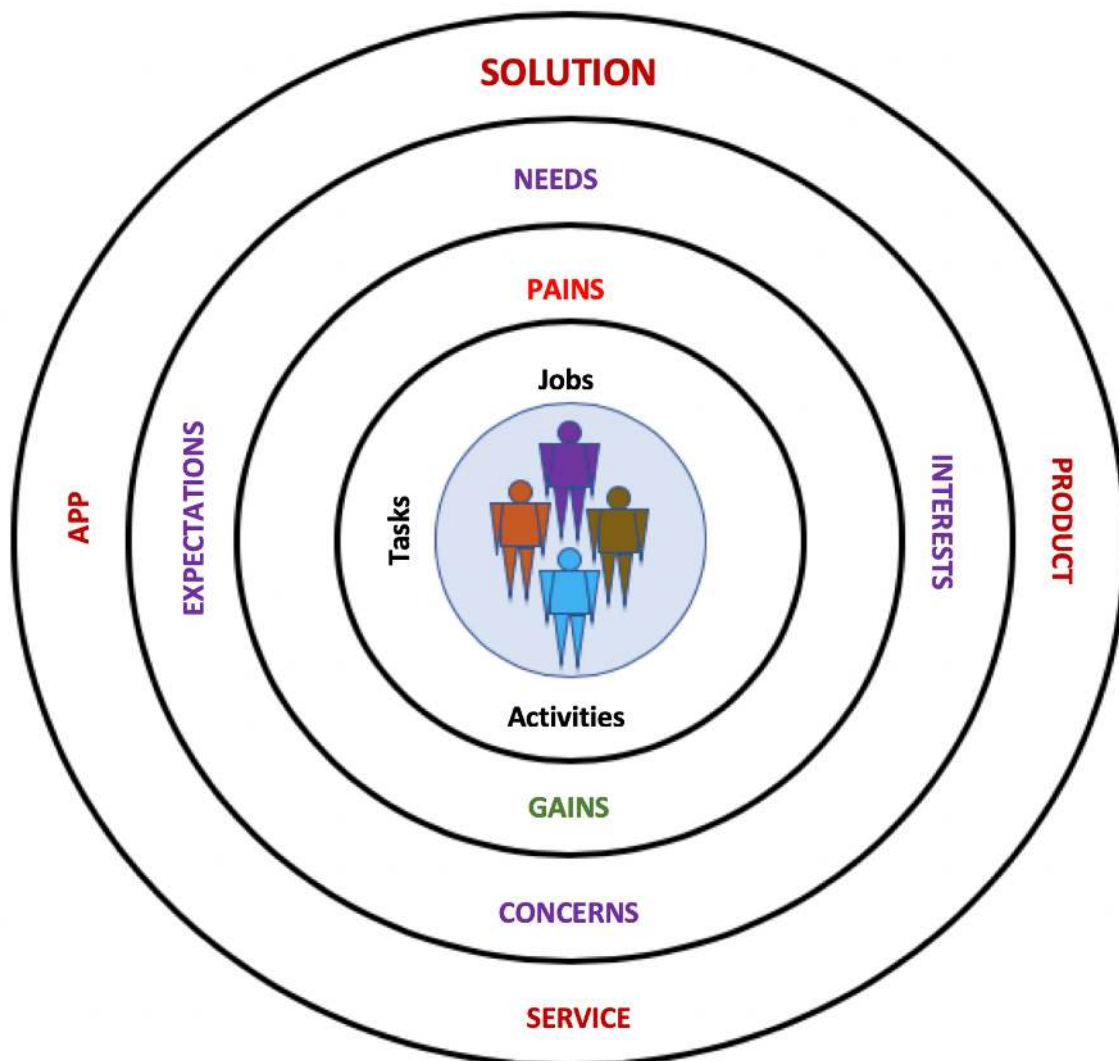
As a Product Owner, you must focus on the usefulness and also the usability by understanding the user experience time to time.

Big picture of Product Owner role



SPACE FOR NOTES

What is Product?



SPACE FOR NOTES



A Product should solve a Particular PROBLEM or Fulfil a specific NEED

PRODUCT LEVEL DESCRIPTION	PRODUCT LEVEL NAME
Basic version of the Product that just fulfil the <i>need</i>	GENERIC
The Product that contains the features that users do <i>expect</i>	EXPECTED
The Product contains the features that are <i>beyond expectations</i>	AUGMENTED
The Product contains the features that are not available yet but may be expected by users in the <i>future</i>	POTENTIAL

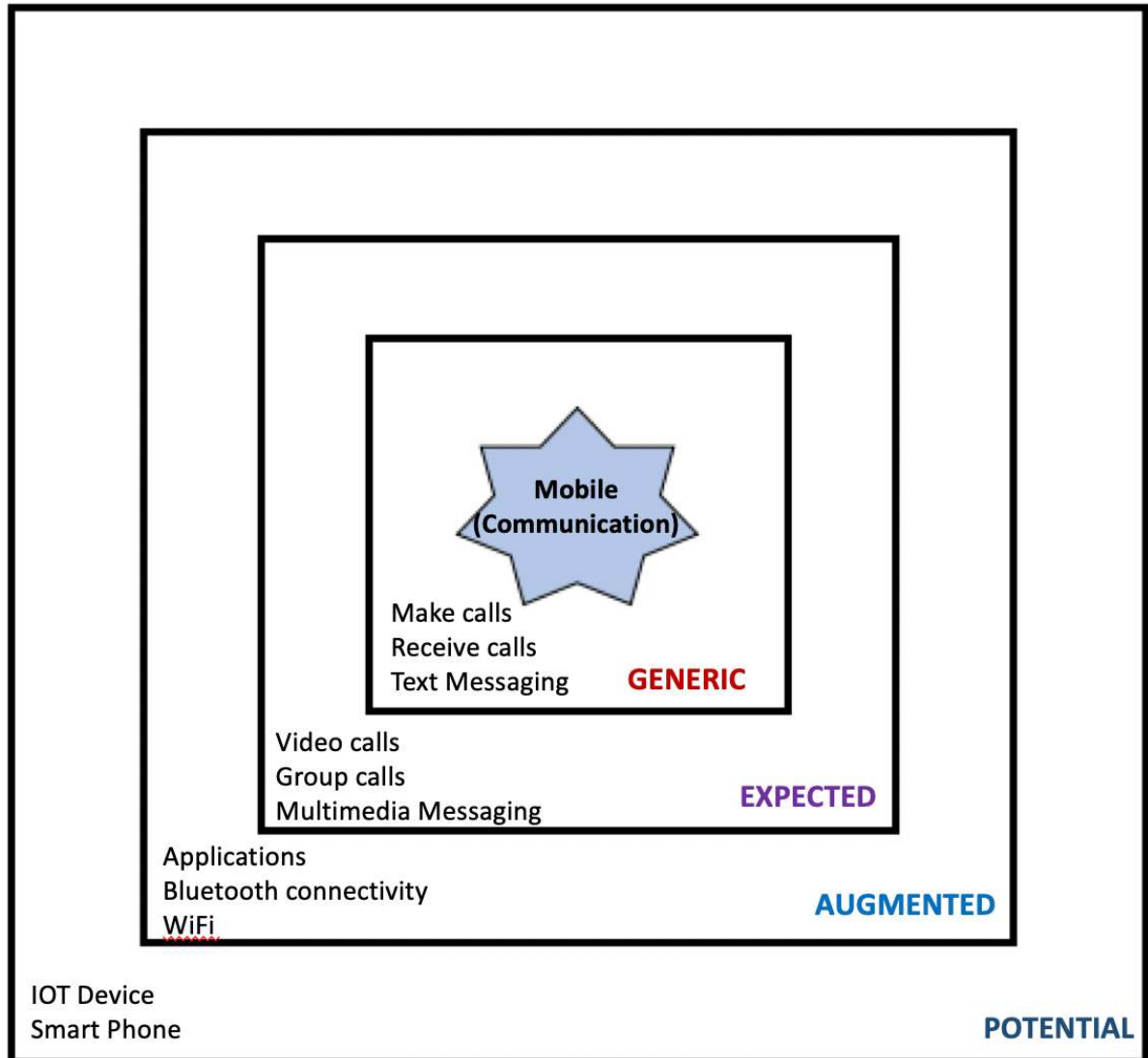
CORONA FACE MASK PRODUCT JOURNEY FROM GENERIC TO POTENTIAL LEVELS

GENERIC	EXPECTED	AUGMENTED	POTENTIAL

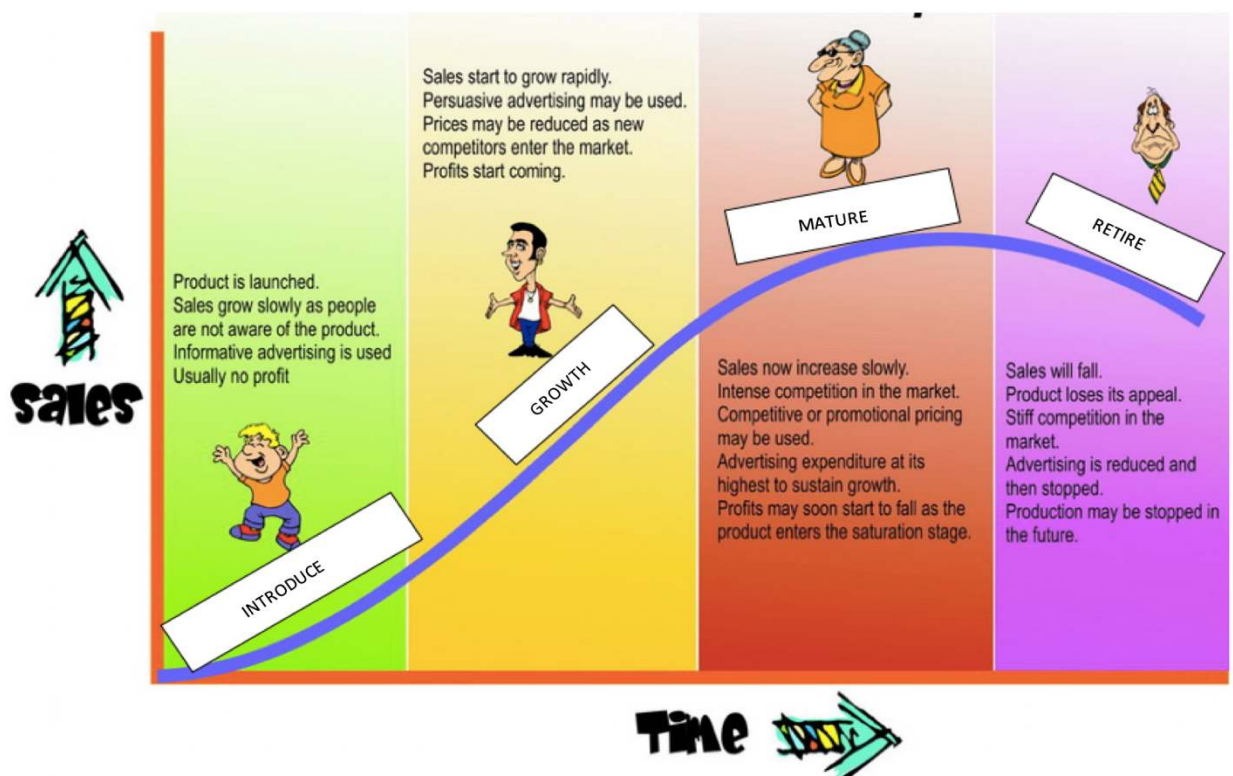
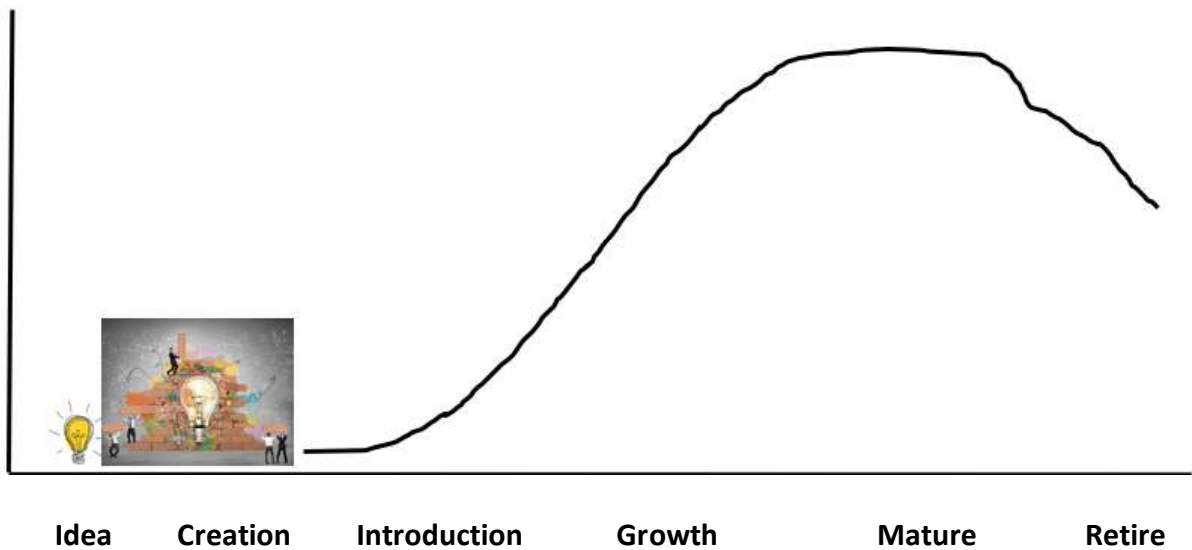
Example:

Product – Mobile

Core benefit: Communication



Product Lifecycle



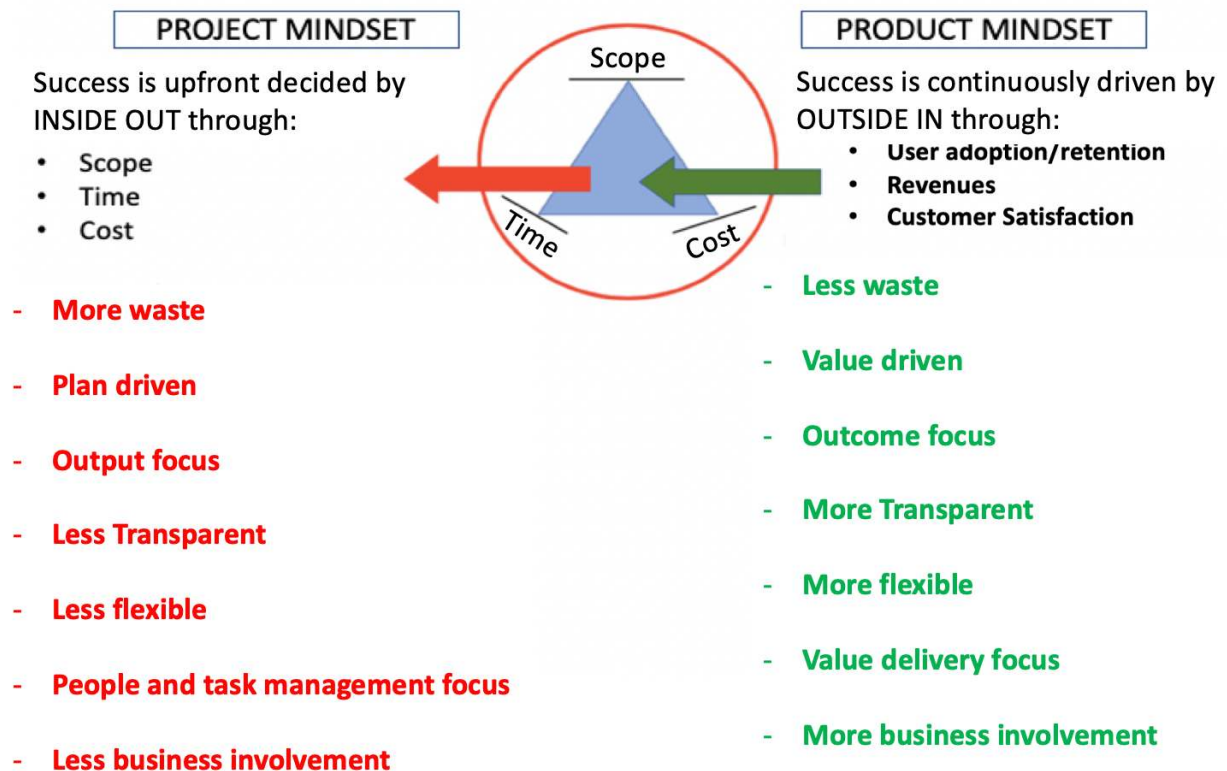
PROJECT Vs PRODUCT

Activity: Understand Project Vs Product

Below are the characteristics of Product and Project.

PROJECT	PRODUCT
TEMPORARY	PERMANENT
MADE FOR ONE CUSTOMER	MADE FOR MARKET SEGMENTS
ONE OFF DELIVERY	CONTINUAL IMPROVEMENTS
PREDICTIVE PLANNING	ADAPTIVE PLANNING
SHORT LIVED PROJECT TEAMS	LONG LIVED FEATURE TEAMS
PROJECT REQUIREMENTS	EVOLVING CUSTOMER NEEDS

PROJECT VS PRODUCT MINDSET

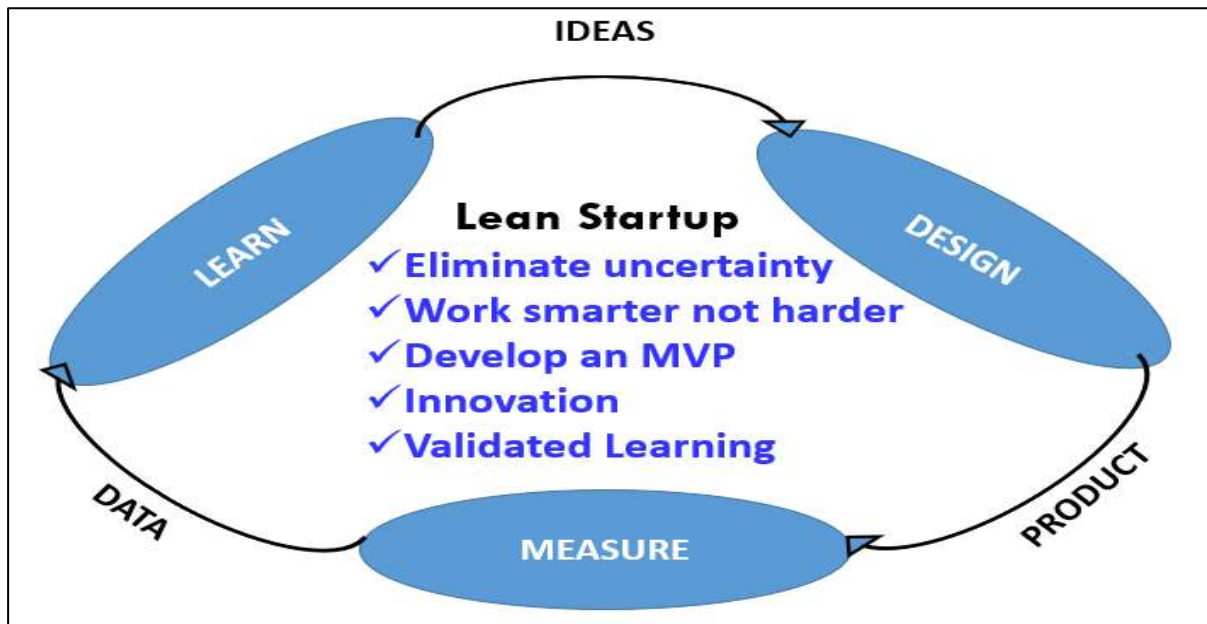


*Shifting from Project
Mindset to Product Mindset
is part of Agile
Transformation*



SPACE FOR NOTES

Minimum Viable Product (MVP)



A minimum viable product (MVP) is a concept from Lean Startup that stresses the impact of learning in new product development. Eric Ries, defined an MVP as that ***version of a new product which allows a team to collect the maximum amount of validated learning about customers with the least effort.***

A key premise behind the idea of MVP is that you produce an actual product (which may be no more than a landing page, or a service with an appearance of automation, but which is fully manual behind the scenes) that you can offer to customers and observe their actual behavior with the product or service. Seeing what people actually do with respect to a product is much more reliable than asking people what they would do.



SPACE FOR NOTES

Examples:



Source: Sam Wong, www.SoupForTheStartupSoul.com



SPACE FOR NOTES

How Airbnb started with MVP?

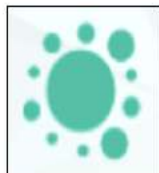
Airbnb started off as a simple site launched by three founders who came up with the assumption that people wouldn't be able to find a place to stay during the Industrial Design Society of America Conference in San Francisco in 2007. So they piled up all their beds in their small apartment and put up a simple ad about renting an air bed on a simple web page. Soon, they had three guests who wanted to share their apartment. Once they had proven that their idea wasn't a waste of time and money, they started looking for how to redesign the platform. Today, Airbnb has 150 million users, 4 million listings, and is valued at \$30 billion. It offers not only accommodation but provides users with lists of restaurants and events in destination cities along with feedback and ratings.

Spotify MVP

In 2006, no one was willing to pay for streaming music as they could easily access it on websites like The Pirate Bay and LimeWire for free. That's when Daniel Ek and Martin Lorentzon had the radical idea of creating a free music streaming service and using ads to generate revenue. But first, they had to test their idea because many other music streaming startups were launched and killed.

So, Ek and Lorentzon did something smart. They created an MVP in the form of a desktop app that offers only one core feature, music streaming. Then, they added the option for customers to pay a monthly fee for an ad-free experience. As they realized the idea had potential, they started enhancing it further. Today, it has a market worth \$18.70 billion.

MVP IS ...



DESIGNED TO CHANGE THE WORLD

By trying new ideas



SUPPOSED TO SOLVE A REAL PROBLEM

It is the problem that makes MVP viable



BUILT TO LEARN USER FEEDBACK

To make it better in the next version



MORE MINIMAL THAN YOU THINK

To learn feedback faster and cheaper way

MVP IS NOT ...



ABOUT GETTING PROFIT

At the first stage its more for learning



DESIGNED TO IMPRESS USERS

Rather, focuses on problem solving



SUPPOSED TO BE AN END PRODUCT

It is a basis for launching new entrepreneurship

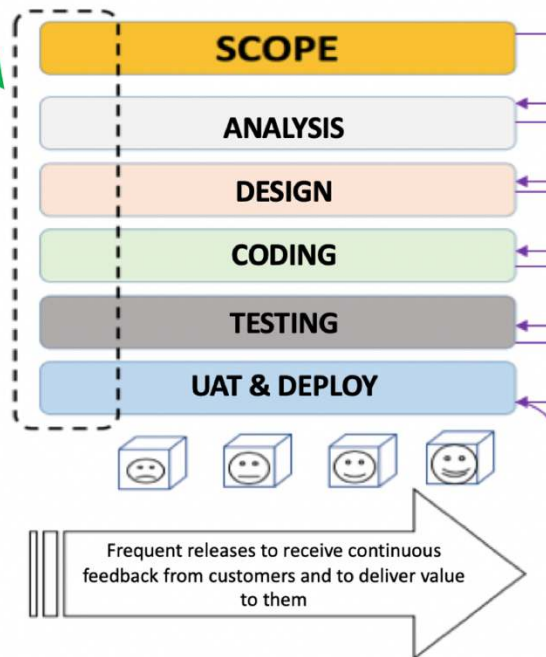


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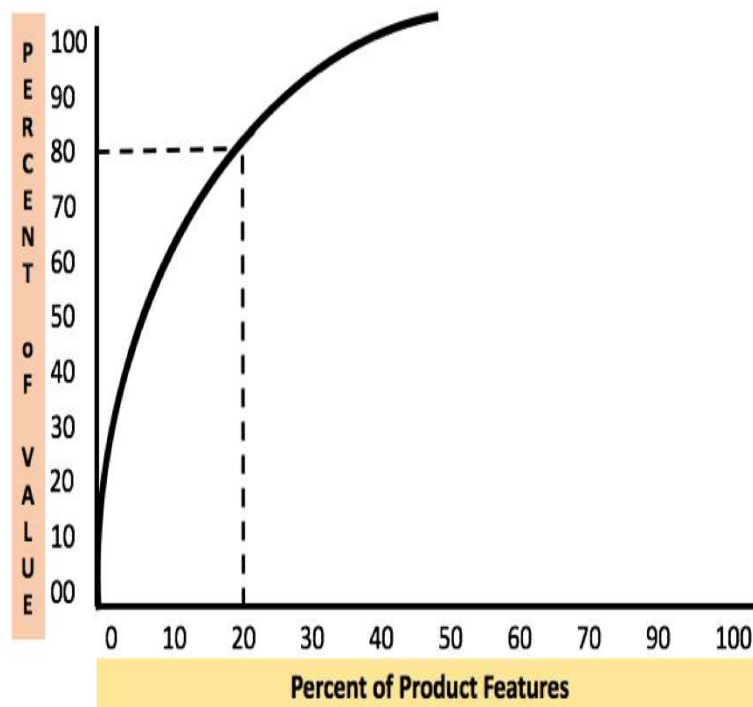
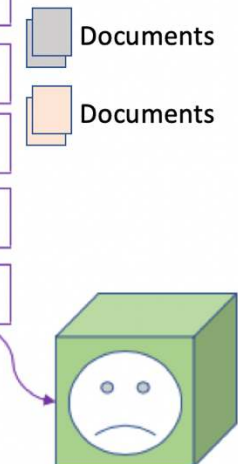
It is supposed to change to get better

Big Bang Vs Iterative & Incremental Delivery

ITERATIVE & INCREMENTAL APPROACH



BIG BANG APPROACH



What is Agility?

Agility is the ability to deliver customer **VALUE frequently and **CONSISTENTLY** while dealing with inherent **UNPREDICTABILITY** and dynamism by recognizing and adapting to **CHANGE****

Frequent delivery helps you to receive early “Feedback”

Consistent delivery helps you to increase “ROI” (Return On Investment)

Why Agile?



Agile Manifesto 4 Values

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and interactions OVER **Processes and tools**
Working software OVER **Comprehensive documentation**
Customer collaboration OVER **Contract negotiation**
Responding to change OVER **Following a plan**

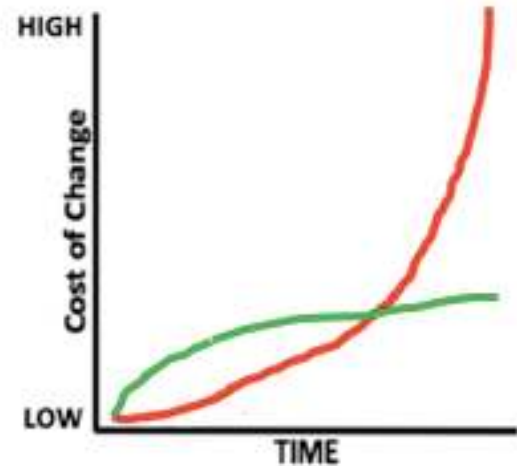
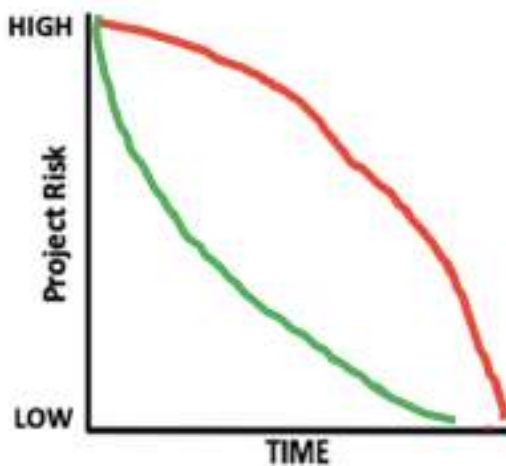
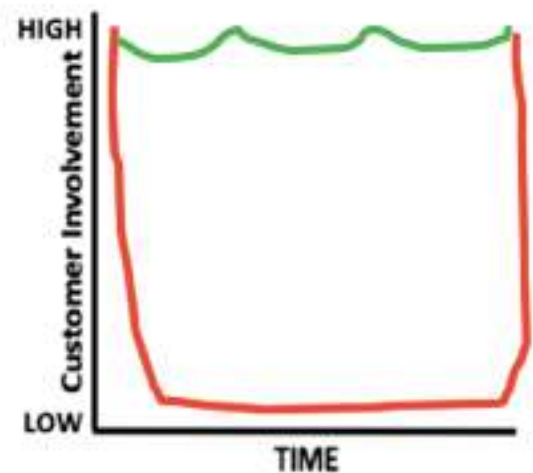
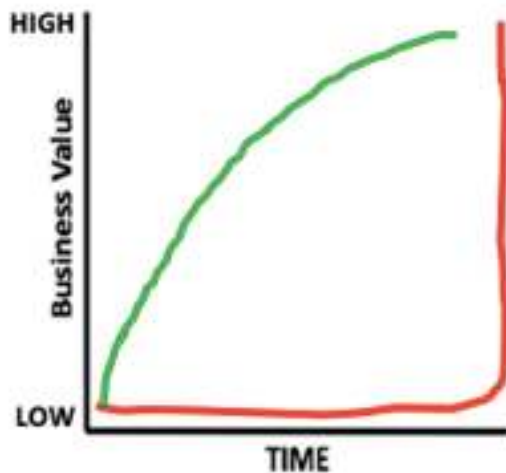
While there is value in the items on the Right side, we value the items on the LEFT MORE!

12 Agile Principles

1. Our highest priority is to satisfy the customer through early and continuous delivery of **valuable** software
2. Welcome **changing** requirements, even late in development. Agile processes harness change for the customer's competitive advantage
3. Deliver working software **frequently**, from a couple of weeks to a couple of months, with a preference to the shorter timescale
4. Business people and developers must work **together** daily throughout the project
5. Build projects around motivated individuals. Give them the environment and support they need, and **trust** them to get the job done
6. The most efficient and effective method of conveying information to and within a Development team is **face-to-face** conversation
7. Working software is the primary **measure** of progress
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a **constant** pace indefinitely
9. Continuous attention to technical excellence and good **design** enhances agility
10. Simplicity--the art of maximizing the amount of **work** not done -- is essential
11. The best architectures, requirements, and designs emerge from **self-organizing** teams
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its **behavior** accordingly

Activity: Understand the below 4 parameters. The **RED** lines indicate the behavior of these parameters in waterfall way of working. **GREEN** lines indicate the Agile way of working.

(2 Minutes)



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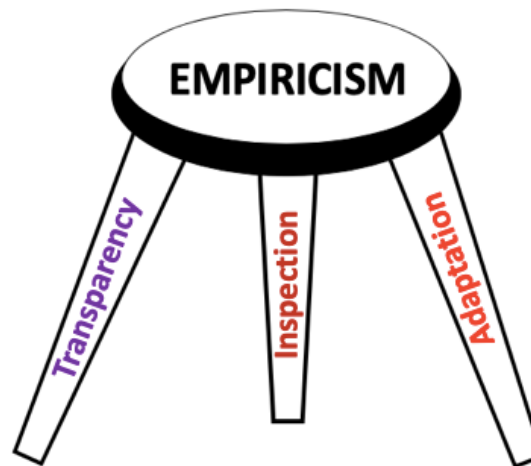
Read more about Agile Values and Principles from the below articles:

4 Agile Values: <https://www.learnovative.com/agile-manifesto/>

12 Agile Principles: <https://www.learnovative.com/agile-principles/>

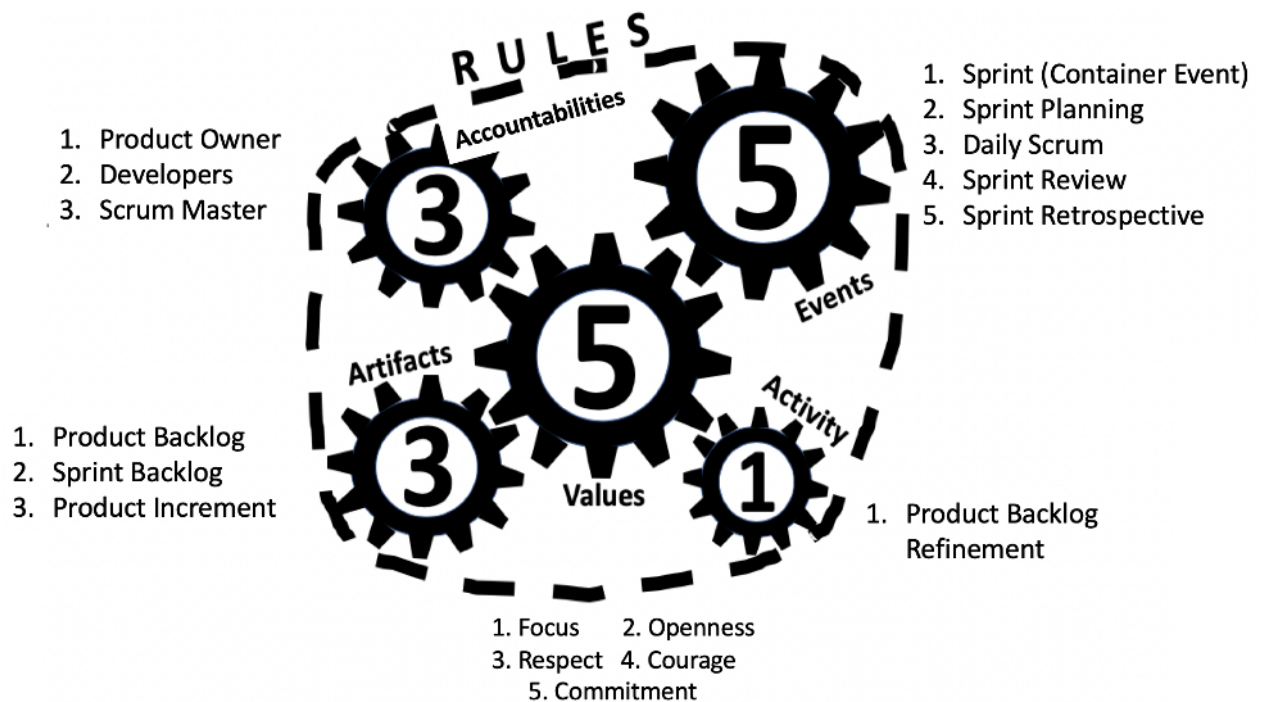
Scrum Framework Overview

- Problems can be different types: Simple & Complex
- Simple means known is more than unknown, complex means unknown is more than known
- Simple Problems can be solved using “Defined” process like waterfall
- Scrum is a **FRAMEWORK** to create adaptive solutions for **COMPLEX** problems
- Foundation for Scrum is **Empiricism**
- Empiricism asserts that knowledge comes from experience and the decisions are made based on what is observed
- THREE legs of Empirical process are:



SPACE FOR NOTES

Scrum Framework Elements



Scrum Team

Product Owner



Developers



Scrum Master



MANAGEMENT

Macro

MANAGEMENT

Micro

MANAGEMENT

Process

MANAGEMENT

EXECUTION

Requirements

DEVELOPMENT

Software

DEVELOPMENT

People & Practices

DEVELOPMENT

:Scrum Team size:

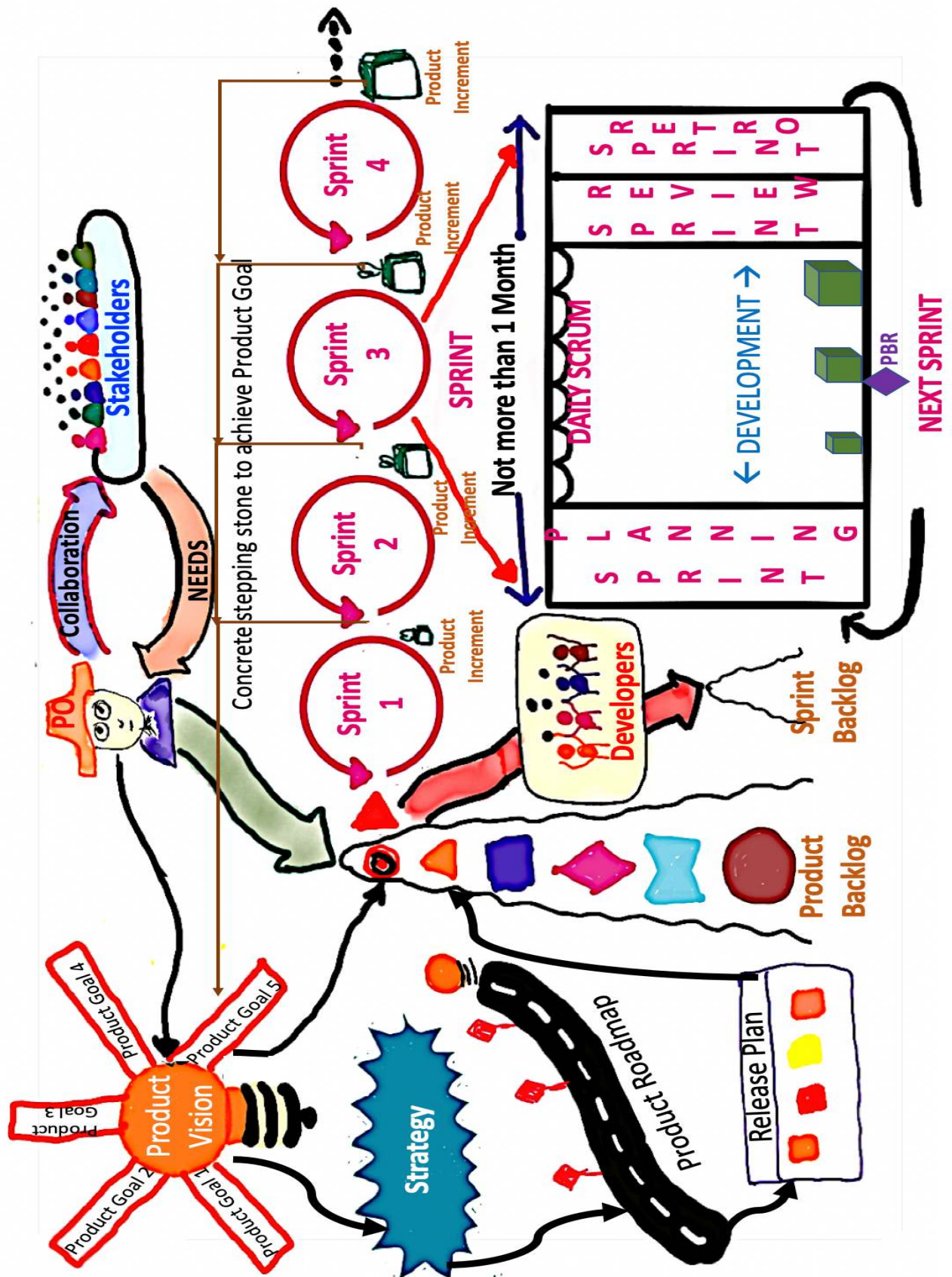
10 or less including Product Owner and Scrum Master
If the team is large, it can be split into multiple smaller Scrum Teams.



SPACE FOR NOTES

Scrum Framework Flow

Activity: Complete the Scrum Framework based on the explanation that we had



Example for Product Vision, Product Goal, Sprint Goal

Become #1 ecommerce product that provides faster, secured, reliable, and seamless shopping experience to our users



Sprint



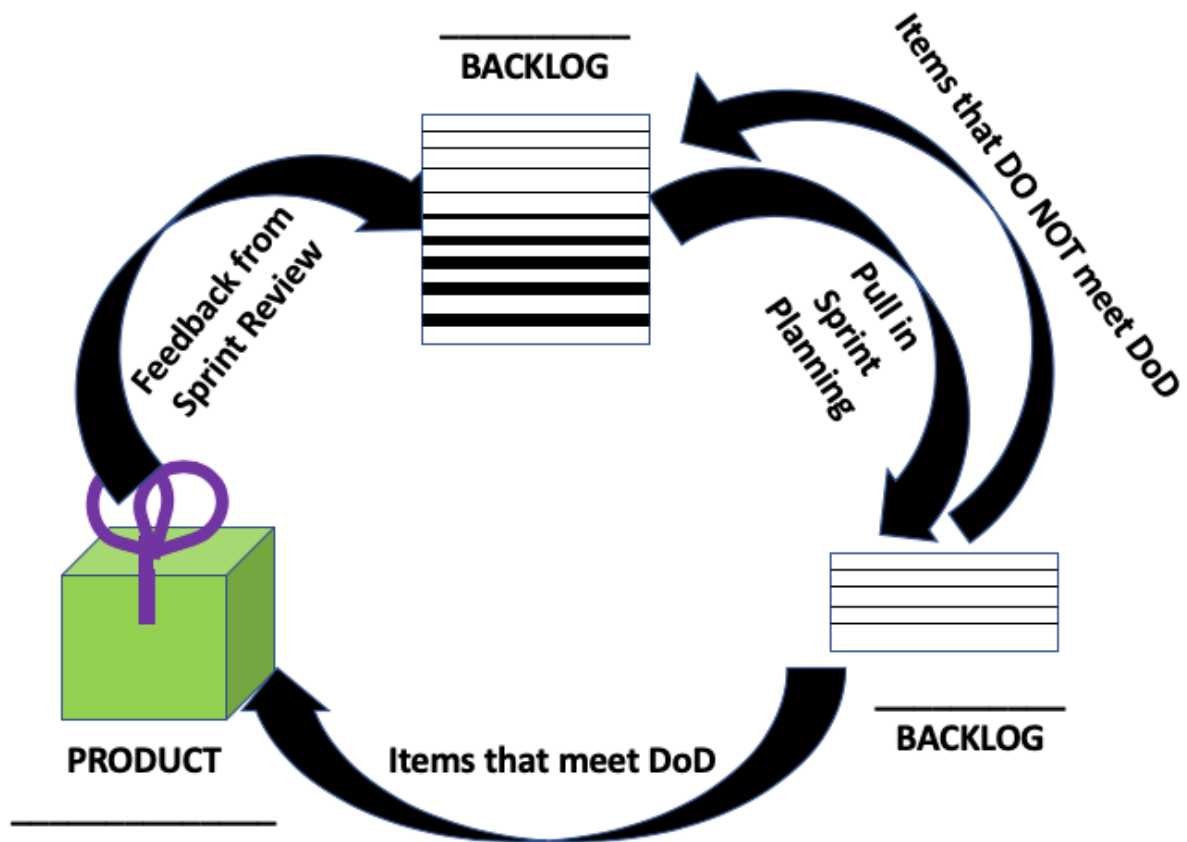
- **Sprint** is the heartbeat of Scrum (a container event)
- It is a container event which contains the other four events
- It is a timeboxed duration with a maximum duration of 1 month or less
- Every Sprint will have 4 events
 - **Sprint Planning:** Maximum duration of 8 hours for a 1 month Sprint
 - **Daily Scrum:** Maximum 15 minutes for any Sprint duration
 - **Sprint Review:** Maximum duration of 4 hours for a 1 month Sprint
 - **Sprint Retrospective:** Maximum duration of 3 hours for a 1 month Sprint
- **Product Backlog Refinement:** Is an ongoing activity to add details such as Description, Size and ORder to the upcoming Product Backlog Items. It can happen anytime during the Sprint between Sprint Planning and Sprint Review.
- **Sprint Characteristics:**
 - Timeboxed
 - Iterative
 - Incremental
 - Protected

Sprint Cancellation

Abnormal termination of the Sprint before Timebox expires is called “Sprint Cancellation”

- Only Product Owner can take call on the Sprint Cancellation
- Only reason for Sprint Cancellation is “When Sprint Goal is obsolete”
- When Product Owner decides to cancel a Sprint, below things happen:
 - Review of the items (if there are any by that time of cancellation)
 - Next Sprint Planning

Scrum Artifacts

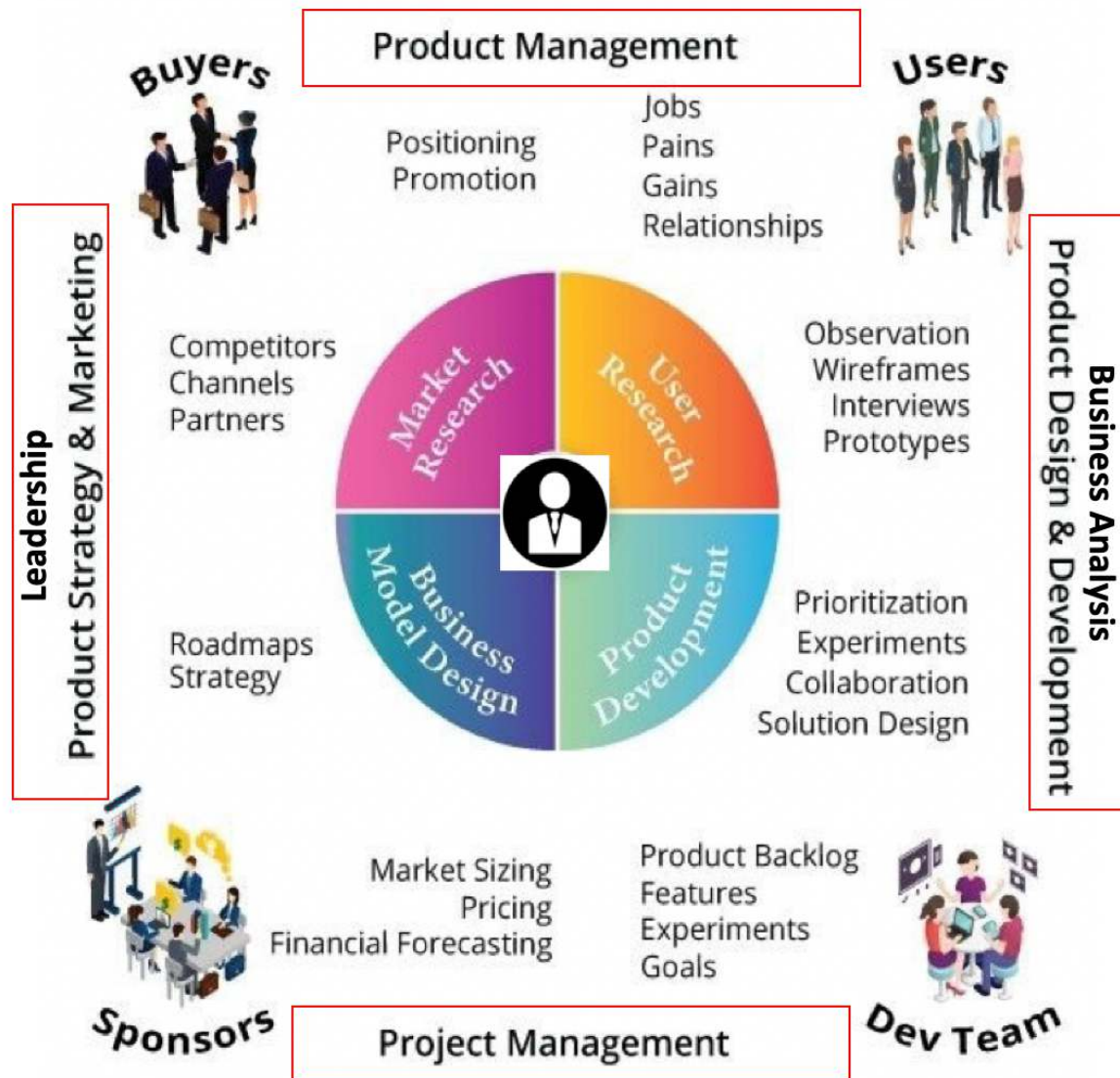


- ✓ Artifacts help to enhance TRANSPARENCY
- ✓ They represent either WORK or VALUE
- ✓ Product Backlog and Sprint Backlog represent WORK
- ✓ Product Increment represents VALUE
- ✓ **Product Backlog's commitment is Product Goal**
- ✓ **Sprint Backlog's commitment is Sprint Goal**
- ✓ **Product Increment's commitment is Definition of Done**



Product Owner Role

Product Ownership is a creative work of solving people's problems or fulfill their needs while strategically helping the business succeed



Product Owner is **one** person, not a group or committee of people. This is because if there are more people acting as Product Owners, the **accountability** will be diluted and it may also lead to **conflicting** with each other and the **decision** making will be delayed or **wrong** decisions will be taken. Primary accountability of the Product Owner is to maximize the **value** of the Product. This may be achieved by **removing** the “no” value items from the **Product** Backlog, keeping the **low** value items in the **bottom** of the product backlog, **market** research, involving the **stakeholders** throughout the product Development to get their **feedback** and so on. Product owner should be **outcome** focused not output. Also, Product owner should understand the customer’s **need** or **problem** and not just the requirements.

The Product owner is also accountable for:

- Developing the communicating the Product **Vision** and **Goal**
- Creating the **Product** Backlog and Product Backlog **Items**
- **Ordering** the Product Backlog **Items**

The Product owner also ensures the Product Backlog is **Transparent** so that everyone who is involved in the Product Development can have shared understanding. Product Owner may **delegate** some of his/her responsibilities to others but he/she remains **accountable** for the consequences.

The Product owner is not a **proxy** between the Stakeholders and the **Developers**. He/she should encourage **direct** interactions between the Stakeholders and the Developers. Product owner tracks the **overall** Product progress using any tools that are suitable. Commonly used tool to track the overall Product Progress is **Burnup** or **Burndown** charts.

The Product owner takes care of the following 4 Ws:

- **WHO?**
- **WHAT?**
- **WHY?**
- **WHEN?**

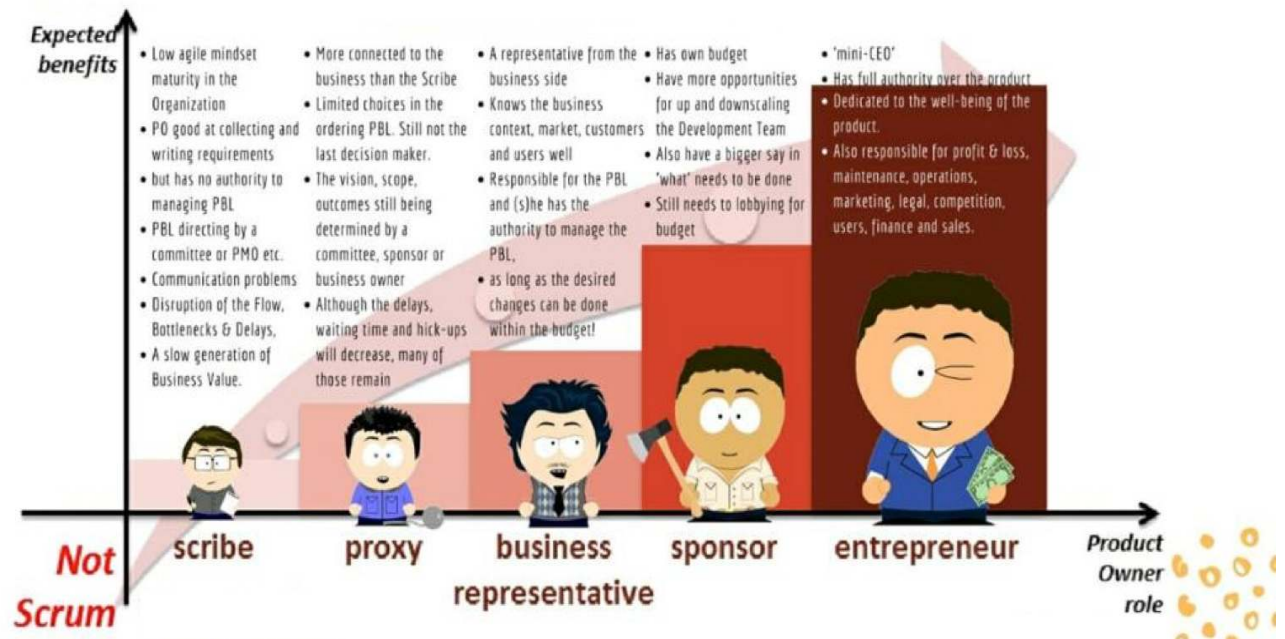
The Product makes some decisions as mentioned below:

- ✓ Content of the **Product** Backlog
- ✓ **Ordering** of the Product Backlog **items**
- ✓ **Accept** or **reject** the work done by Developers
- ✓ When to release the Product **increment** to stakeholders
- ✓ When to **cancel** a Sprint

Product Owner Activities

PRODUCT OWNER ACTIVITY DESCRIPTION
Creating and communicating the Product Vision and Goals
Responsible for content, availability and order of the Product Backlog
Maximising the value of the Product
Create requirements from the needs/wants/desires of stakeholders
Creating Product Roadmap
Ensuring that the Product Backlog is transparent, visible and understood
Manage stakeholder expectations
Provides answers to the queries of the Developers on backlog items
Leading the Release Planning
Finding strategies for stakeholder management
Managing Product Backlog
Keeps the Product inline with the Business Objectives
Participates in Sprint Review event
Make sure the Product Goal is understood by Stakeholders and the Scrum team
Tracks overall Product or Release progress
Leading the product backlog refinement activity
Conflict resolution between stakeholders with respect to their expectations
Participates in Sprint Retrospective event
User & Market research
Competitor product analysis
Decides when to release the increment to the users
Identify and manage the Functional dependencies through Product Backlog
Participates in Sprint Planning event
Cancels the Sprint when the Sprint Goal is obsolete
Economic conditions observation and analyse impact on the Product
Participate in the Retrospective
Collaborates in creating the Sprint Goal
Reviews and provides approval/feedback for the items done by Developers during Sprint
Updating the Sprint Review feedback in the Product Backlog
Making the Scope Vs Time Trade-off decisions

Product Owner Maturity Levels



First 3 are referred as “Receivers” because they work based on instructions

Last 2 are referred as “Initiators” because they take decisions

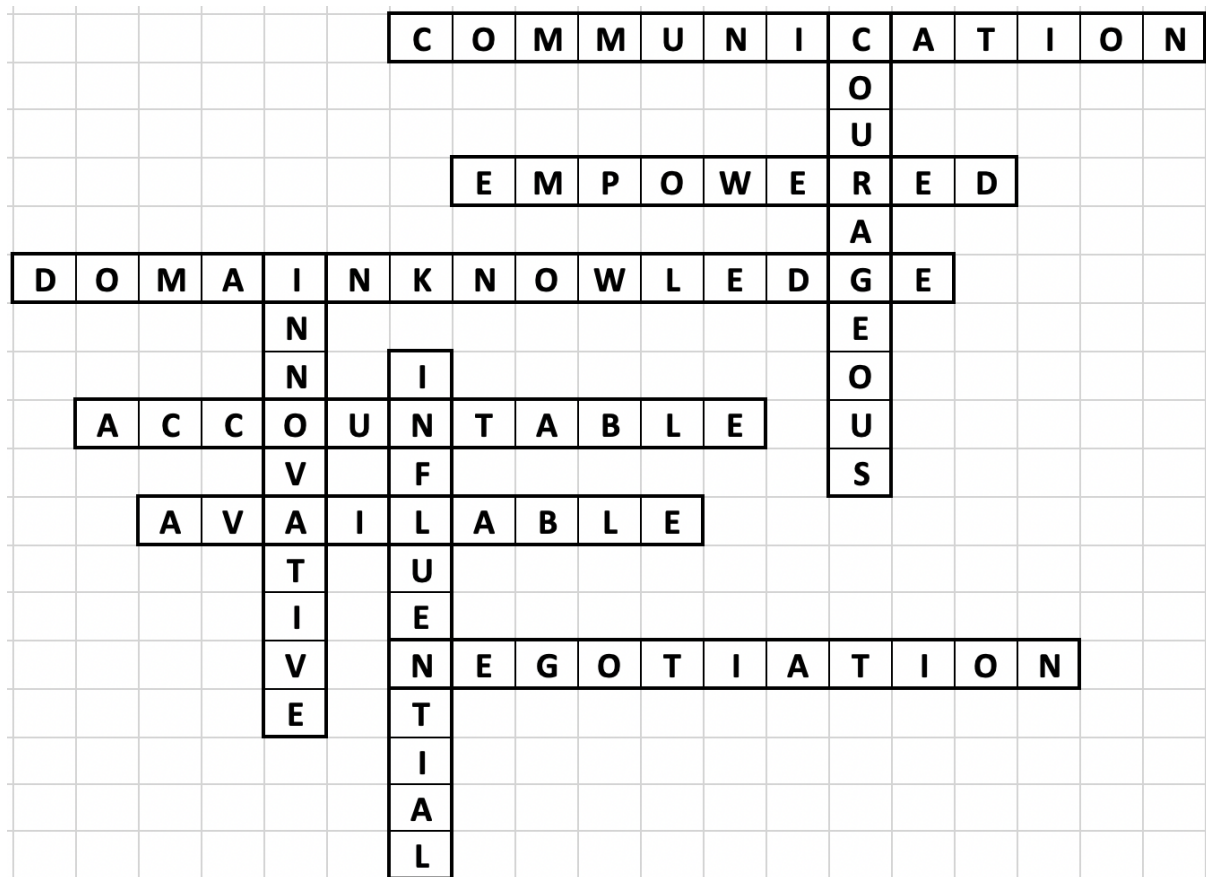
Entrepreneur is the Real Product Owner

Product Owner Role In A Sprint



S No	Activities of Product Owner	Respective part of Sprint
1	- Represents the Stakeholders to Developers - Explains/Presents the Backlog items to the Developers - Collaborates with Developers to formulate the Sprint Goal - Provides clarifications - Provide Trade-off inputs	Sprint Planning
2	- Provides his inputs to the Developers on anything related to Process/Tools/Relationships and other areas - Takes feedback from the team on any improvements that Product owner has to work up on	Sprint Retrospective
3	- Product owner collaborates with Developers for this activity - Involves the Developers to get their support in estimates	Product Backlog Refinement
4	- Available to the Developers on need basis - Provide clarifications - Verifies the items that are done and provide inputs - Approves the items that are done	Development
5	- Represents the Developers to the Stakeholders - Explains what is done/not done - Works with stakeholders to get their feedback - Updates stakeholders on the forecast/Market conditions - Together with stakeholders and Dev & Scrum Master discusses that needs to be done next Sprint	Sprint Review
6	- Product owner is optional - If Developers wants, then Product owner should attend - Purpose is to provide clarifications	Daily Scrum

Product Owner Key Characteristics



1. What if there is not Product Owner assigned to a Product?

It will be like a ship without a Captain. The Product leadership and direction will be missing in this situation and also the Developers will be bombarded with the Stakeholder requests without having anyone to prioritize. If the Product is a failure, there will be no one to own the accountability.

2. What happens if you relabel a Project Manager as Product Owner?

Traditional Project Manager is more closer to task management, people management, coordination and output focused whereas Product Owner is the Value Maximizer and outcome focused. So without a proper transformation if simply a Project Manager is relabeled as Product Owner things will not change much from a traditional way to Agile.

3. Should Product Owner be technical expert?

Not necessary but it will be a added advantage for the Product Owner if he/she has a high level technical knowledge so that the conversations with the Developers will be faster. The Product Owner's primary focus is on Why, What but not How.

4. Is it good to have onsite and offshore Product Owners separately?

No, it will dilutes the accountability and delays the decisions.

5. What are the challenges for Product Owner if there are more Scrum Teams working and how they can be addressed?

Availability, Clarifications to different Scrum Teams throughout the Sprint, validation of the work done by each Scrum Team throughout the Sprint and Time management, keeping enough work to engage all the teams.

Joint Scrum events like Joint Sprint Planning, Joint Sprint Review, Joint Sprint Retrospective and the activity Joint Product Backlog Refinement will help to address these challenges.

Project Manager Vs Product Owner

PM Characteristics	PO Characteristics
Believes and depends up on structure and authority.	Believs and relies and creativity and collaboration
May influence Product Development depending up on the resource availability.	Understands the business and also the user experience
Tracks the progress based on milestones.	Tracks progress through value and working software
Works based on project plan and focus on deadlines.	Works through prioritization
Focus is in resources, scope, time, cost.	Focuses is on customer, markets, revenue etd
Owens "HOW" and "WHEN".	Owens "WHO", "WHAT", "WHY", "WHEN"
Approach is based on the "PLAN" driven delivery.	Approach is based on "VALUE" driven delivery
Believes in control and micromanages the people working on the Product.	Owens and manages the Product Vision, Strategym Roadmap
Delivers Project based on the requirements received from the customers.	Delivers Product by understanding the needs, or problem of the customers/users

Read below Articles for more insights about Product Owner:

<https://www.learnovative.com/scrum-product-owners-characteristics-rights-and-responsibilities/>

<https://www.learnovative.com/product-manager-vs-product-owner/>

<https://www.learnovative.com/common-challenges-for-product-owners/>

<https://www.learnovative.com/why-product-owner-and-scrum-master-cannot-be-the-same-person/>

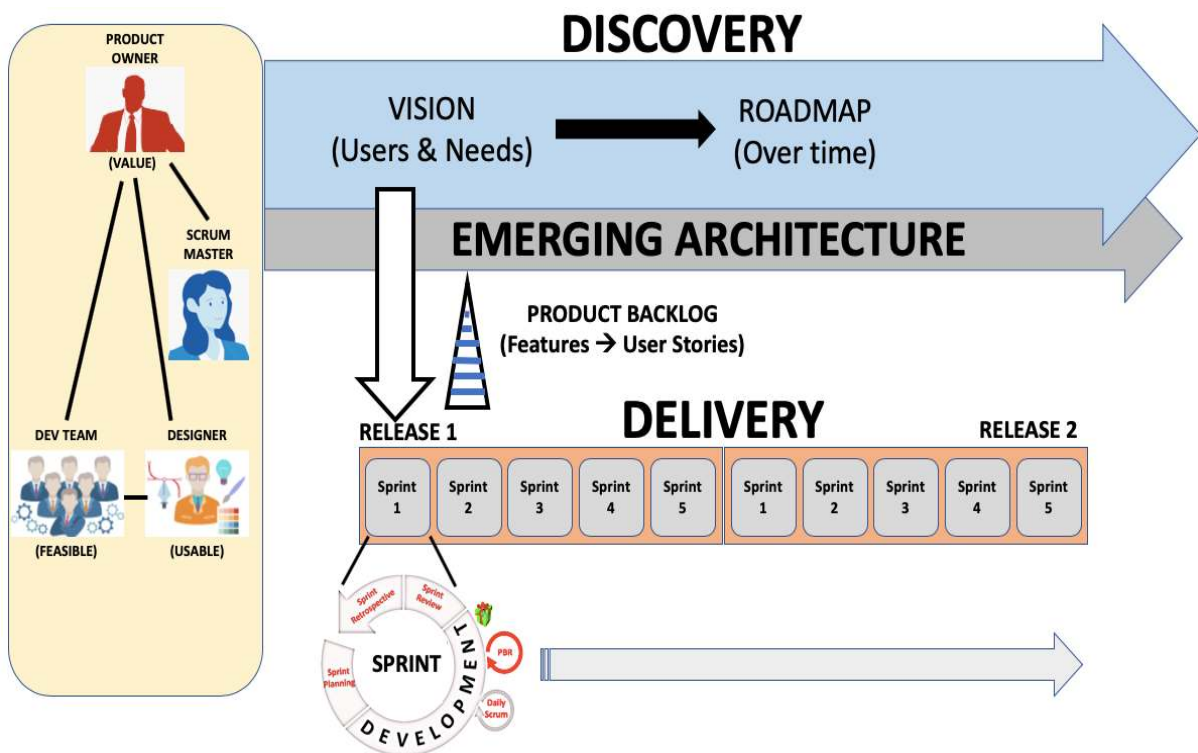
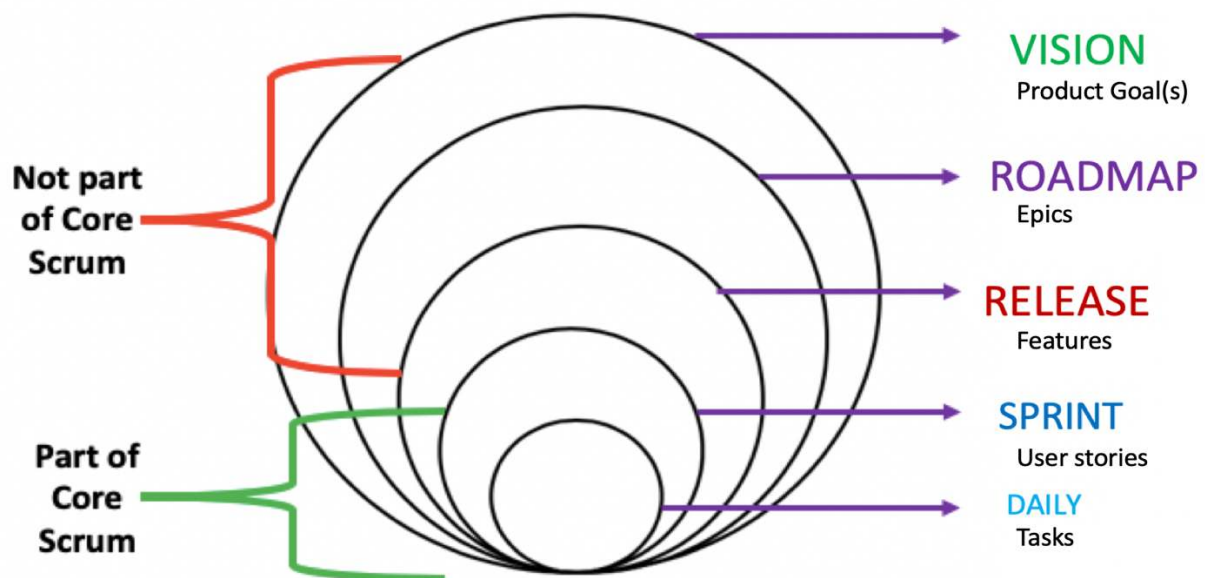
<https://www.learnovative.com/different-shades-of-the-product-owner-role-in-organizations/>

User experience is important



Agile Product Planning (The Big Picture)

Planning is everything, Plans are nothing!



CERTIFIED SCRUM PRODUCT OWNER (CSPO) WORKSHOP

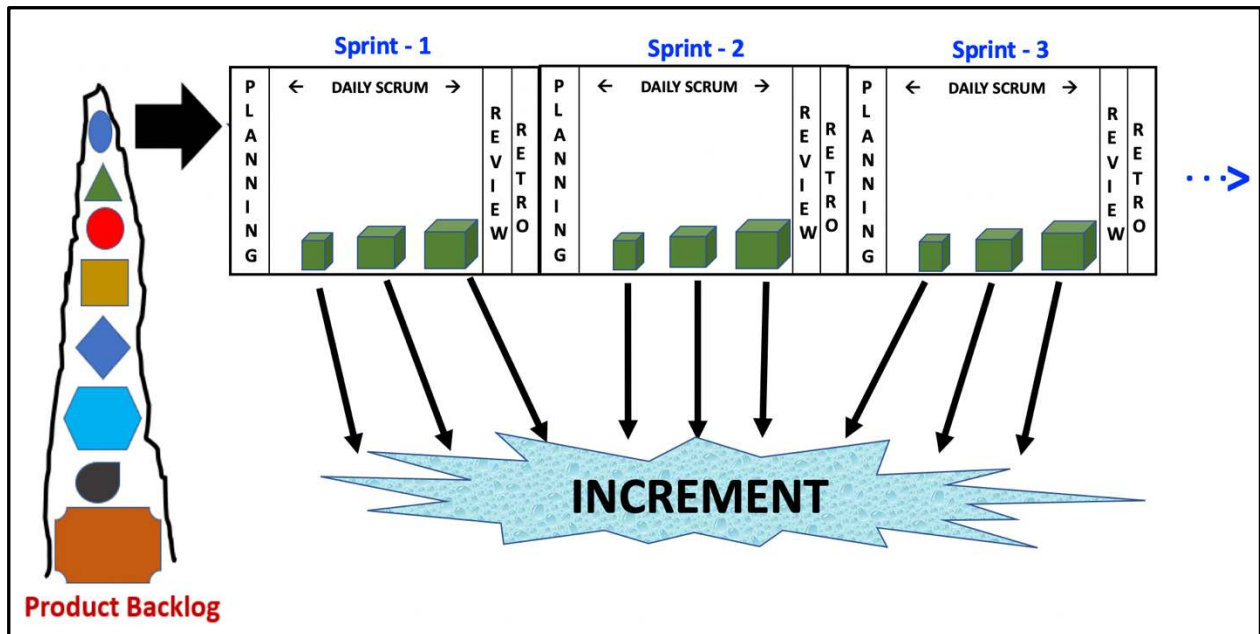
From the above planning levels, discuss and fill the following:

Planning Level	Part of Scrum?	How to represent?	Who creates	Review Frequency
Vision	No	Elevator Statement Vision Poster Press Release One Pager	Product Owner (Either creates or co-creates)	Yearly once
Roadmap	No	Roadmap contains 2 to 3 Releases	Product Owner	Twice an year
Release	No	Release Plan	Product Owner with the help of Team(s)	Quarterly
Sprint	Yes	Sprint Goal & Sprint Backlog	Scrum Team	As per Sprint duration (1 – 4 weeks)
Daily	Yes	Sprint Backlog	Developers	Daily



SPACE FOR NOTES

INCREMENT



- Developers are committed to create the Increment
- Increment is the collection of the items completed during the Sprint that meet the Definition of Done
- The increment's commitment is "Definition of Done"
- The increment must be in a usable condition irrespective of Product Owner wants to release or not
- Is a small step towards the "Product Goal"
- There can be more than one increment in a Sprint
- Whenever a Product Backlog Item meets the Definition of Done, an increment is born

Reasons for releasing:

Customer Request
Market Opportunity
Required Release
Competitive Response
Major Release
Maintenance

Definition of Done

- ✓ It is a commitment for the Increment
- ✓ Quality criteria checklist
- ✓ Helps Product Owner and Developers have a shared understanding on “Done”
- ✓ Created by Scrum team
- ✓ Different for different products
- ✓ Created before Sprint planning
- ✓ If there are any organization level quality standards exist, they must be part of DoD
- ✓ Helps Developers in Sprint planning to forecast the work
- ✓ In multi-team environment, all teams should have a common Definition of Done
- ✓ Sprint retrospective is a formal opportunity for Scrum team to inspect and adapt

ACCEPTANCE CRITERIA	DEFINITION OF DONE
Related to functionality	Related to quality
Owned by Product Owner	Owned by Developers
Specific to each Product Backlog Item	It is a commitment to increment
Can be created at any time before accepting into a Sprint	Usually defined before the first Sprint and inspected and adapted as needed in the Sprint Retrospective



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Read more about Definition of Done below:

<https://www.learnovative.com/definition-of-done/>

Product Discovery

Do you know the Top 10 reasons for start-ups to fail?



Source: <https://www.strategyzer.com/blog/posts/2018/10/15/why-companies-work-on-products-nobody-wants>



SPACE FOR NOTES

Product Discovery is the process that helps the Product Owner to identify ideas to build Products that are **USEFUL** and also **USABLE**

What is the difference between a User and Customer?

User: Who uses the Product without paying

Customer: Who uses the Product by paying

How do you empathize with Users as a Product owner?

“User Research” is one of the powerful ways to understand use behaviors, their needs, pains and gains through some observations techniques. Key to user research is “Empathy”. Empathy is to understand user’s needs, desires, and wants from their point of view.

User Research for a new Product is different from User Research for an existing product. For a new product (0 → 1 Stage), User Research is done to validate product ideas and hypotheses. For an existing product (1 → n Stage), User Research is done to get feedback on features of the product.



User research helps in:

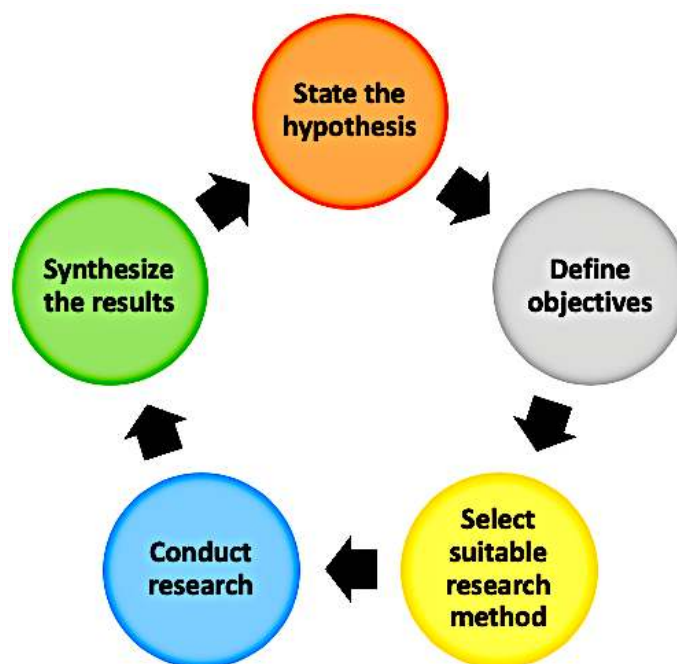
- ✓ Validate assumptions and hypothesis
- ✓ Get qualitative data about user needs/desires/behaviors
- ✓ Understand how users will interact and use your product
- ✓ Promotes iterative approach to refine your product
- ✓ Gives in depth understanding and clarity on what works and what does not work in your Product



User Research helps improving the Design:

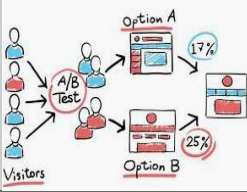
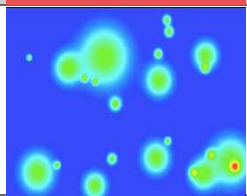


Steps to conduct User Research:



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Common methods to conduct the User Research:

Description of the User Research Method	
	A/B Testing: This is a quantitative method in which you can create two versions of a product or feature, and evaluate which one works better with the users
	Contextual Inquiry: Users are observed performing tasks in their original habitat to gather information about the context of use of the product
	Focus Group: Moderated discussion with a group of users, allow you to learn about user attitudes, ideas, and desires
	Heatmap: A heatmap is a graphical representation of data where values are depicted by color. Heatmaps make it easy to visualize complex data of users and understand it at a glance
	Interview: These are one-on-one discussions with individual users to gather qualitative data about their pain points and motivations
	Prototype: Allows the design team to explore ideas before implementing them by creating a mock-up of the site. A prototype can range from a paper mock-up to interactive html pages
	Survey: Consists of a questionnaire with questions and structured responses, and comes in handy when you need to quantify qualitative information from users
	User Feedback: You can ask users to give their opinions on the product — their likes and dislikes, and if they would recommend the product to others

How to select suitable User Research method?

Each User Research method has its own advantages and drawbacks, and it's important to understand which method to use depending on your product, your users, and the kind of information you want to gather — information on what users think or what they do.

If you want to understand what users say or think, surveys, interviews and focus groups can help you gauge their beliefs and attitudes. If, on the other hand, you want to focus on what users do or how they use your product, methods like A/B testing and heatmaps are more suitable.

Interviews are suitable for gathering qualitative data, whereas surveys are handier for collecting quantitative data.

Select most appropriate User Research Technique for the below examples

Example scenario description	Suitable User Research Method
Your users are scattered across different locations and you do not have time to meet everyone in person. But you need to understand their needs.	Survey
Your product is already into market and your users are using it. You want to identify which features have issues with usability or performance issues based on the information you received from your customer support team.	Heatmap
You have a new feature in your Product which has a complex navigation and it is also longer. So how do you get inputs from users before implementing the feature to make sure the navigation is correct?	Prototype
You are planning your next major release planning. For that you would like to get critical information from the users who are already using the current version of your Product. How do you collect the feedback in this scenario?	Interview User Feedback Focus Group
One of the critical stakeholders has a language problem hence she cannot explain you the requirement of the Product you are building. However she is very critical stakeholder so how do you empathize with this Stakeholder?	Contextual enquiry
You are a Product Owner for one of your company's internal Product and almost all your company employees will use this new Product. All your employees are collocated so what is the best approach to collect the inputs?	Interview/Focus Group/Survey (Based on # of people to cover)

Persona

- ✓ Fictional character who interacts with the Product/Service/Result
- ✓ Useful in considering the goals, desires, concerns
- ✓ Helps to identify the features based on different Personas
- ✓ Helps in:
 - Coming up with better user experience
 - Prevent some common design pitfalls
 - Avoid “self-referential” designs
- ✓ Persona is key in building products that are close to customers’ needs

How to capture the Persona details?

Use Persona Canvas suggested by Roman Pichler. Explore this approach to get more information.

Example Persona Canvas:

	• Male • 25 Year • Education – Engineering in Computer Science • Profession – Software Developer	• Location – Mumbai • Device – iPhone, MacBook
Characteristics/Personality: • Outgoing • Extrovert • Frequently go out on parties • Intelligent • Sports - Football	Interests: • Coding books • Sports Fan • Manchester United	Needs • Technology Magazines • Latest Apple Gadgets • Manchester United Merchandise



SPACE FOR NOTES

Value Proposition Canvas

The Value Proposition Canvas is a tool which can help ensure that a product or service is positioned around what the customer values and needs.

The Value Proposition Canvas was initially developed by Dr Alexander Osterwalder as a framework to ensure that there is a fit between the product and market. It is a detailed look at the relationship between two parts of the Osterwalder's broader Business Model Canvas; customer segments and value propositions.

The Value Proposition Canvas can be used when there is need to refine an existing product or service offering or where a new offering is being developed from scratch.

The Value Proposition Canvas is formed around two building blocks – customer profile and a company's value proposition.

Customer Profile

- Customer jobs – the functional, social and emotional tasks customers are trying to perform, problems they are trying to solve and needs they wish to satisfy.
- Gains – the benefits which the customer expects and needs, what would delight customers and the things which may increase likelihood of adopting a value proposition.
- Pains – the negative experiences, emotions and risks that the customer experiences in the process of getting the job done.

A customer profile should be created for each customer segment, as each segment has distinct gains, pains and jobs.

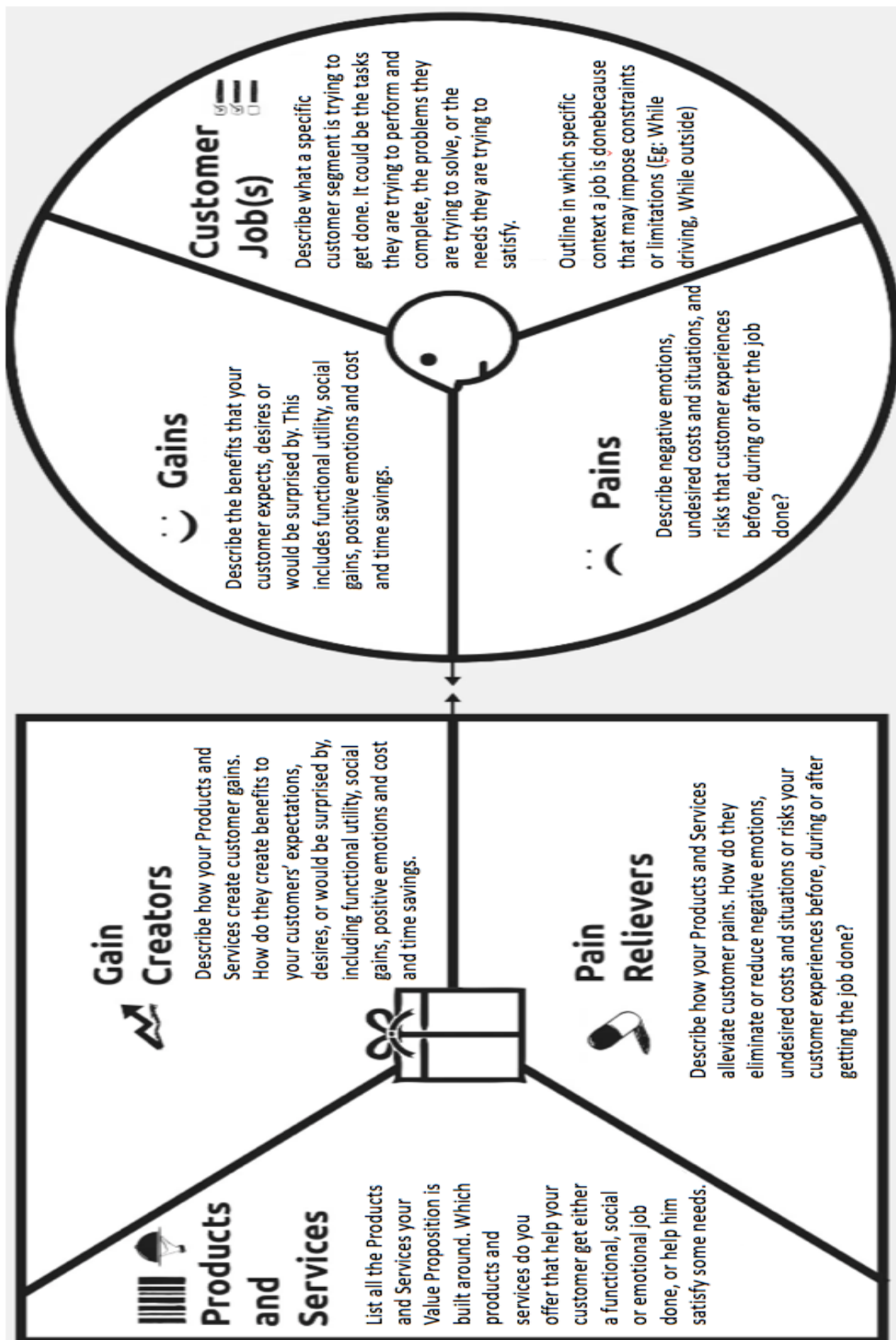
Value Map

- Products and services – the products and services which create gain and relieve pain, and which underpin the creation of value for the customer.
- Gain creators – how the product or service creates customer gains and how it offers added value to the customer.
- Pain relievers – a description of exactly how the product or service alleviates customer pains.

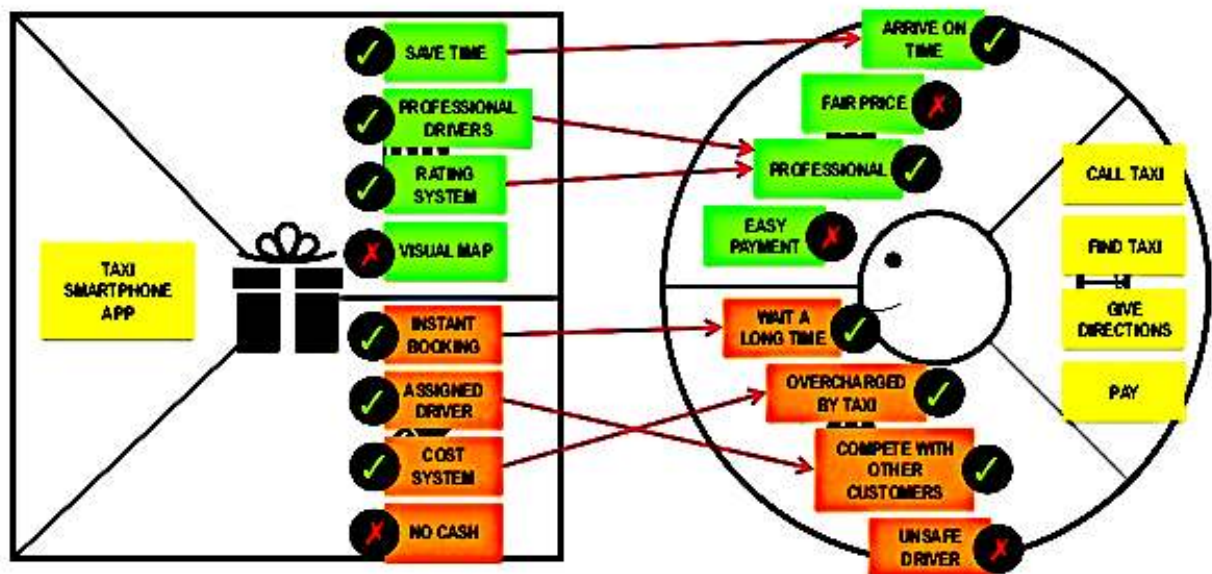
Achieving fit between the value proposition and customer profile

After listing gain creators, pain relievers and products and services, each point identified can be ranked from nice to have to essential in terms of value to the customer. A fit is achieved when the products and services offered as part of the value proposition address the most significant pains and gains from the customer profile.

Identifying the value proposition on paper is only the first stage. It is then necessary to validate what is important to customers and get their feedback on the value proposition. These insights can then be used to go back and continually refine the proposition.



Example: Value Proposition Canvas for “A Taxi Smartphone”.



Product Vision



VISION without **ACTION**
is a *Daydream*, but
ACTION without **VISION**
is a *Nightmare*

VISION IS THE ART OF SEEING THINGS INVISIBLE

- Jonathan Swift

Product Vision helps you to “BEGIN with the END in the mind”

Product Vision indicates two factors.
Why the Product is being Built and

What is the desired end state

Vision covers the following:

- Who will buy your product?
- Who is target Customers?
- What Problem/need the Product will address?
- What are the critical success factors?
- What are unique differentiating factors?

A Good Product Vision should be:

BIG:

Product Vision should be Big/Broad so that you may have multiple strategies to meet the Vision. If the Vision is Narrow then there may be only very few strategies to meet the Vision.

INSPIRING:

Product Vision should inspire people who are going to use it so that it motivates them to buy the Product.

CONCISE:

Product Vision should be crisp and simple to understand so that it can be easily explained and understood.

SHARDED & STABLE:

Product Vision should be shared across all the people involved in the Product Development (Users, Customers, Stakeholders, Scrum Team(s)) so that most of the decisions can be taken by respective people. Product Vision should be stable so that it provides the direction and also helps to validate the path to success.

Most popular Vision Statements:



To create economic opportunity for every member of the global workforce.



Tap a button, get a ride.



To organize the world's information and make it universally accessible and useful.



Book rooms with locals, rather than hotels.



To provide the best in cosmetics innovation to women and men around the world with respect for their diversity.



Make it easy for anyone to create a beautiful and powerful online store.

Advantages of Powerful Product Vision:

- Vision provides direction and helps the organization prepare for the future.
- Vision provides guidance for decision-making.
- Vision can help to decide what you will and what you will not do.
- Vision helps set priorities and guides planning.
- Vision aligns people and activities across the organization.
- Vision provides purpose and a source of inspiration.
- Vision empowers people and helps focus their efforts.

Product Vision can be created using following techniques:

- **Elevator Statement**
- **Product Vision Box/Poster**
- One Pager
- Press Release

Elevator Statement

FOR <Target Customers>
WHO <Statement of Their Need>
THE <Product Name> **IS A** <Product Category>
THAT <Key benefits, Compelling reasons to buy>
UNLIKE <Primary competitor/Alternate Product>
OUR PRODUCT <Unique differentiation Factor>

Create a Vision using Elevator Statement for the Product you identified. Each table as a group and read it to let others know



Timebox: 5 Minutes

Example Elevator Statement:

FOR < Those are into Product Management >

WHO < Wants to manage their Products better and effective >

THE < CSPO > **IS A** < Scrum Alliance Certification >

THAT < makes you standing out of the crowd >

UNLIKE < Traditional output focused Project Management
Certifications >

OUR PRODUCT < Helps you to become outcome focused Product
Owners and build customer centric Products >

Product Vision Board

Below is a sample Product Vision board for a Rental platform.

 VISION What is the reason for creating the product? What positive change should it create?		Making the long-term home rental process for tenants and landlords simple, faster, easier and global Will save plenty of time for both tenants and landlords while establish trust between them	
 TARGET GROUP Which market or market segment does the product address? Who are the target customers and users?	 NEEDS What problem does the product solve or which benefit does it offer? If you identify several needs, prioritise them and move the most important one to the top.	 PRODUCT What product is it? What are its three to five stand-out features that set it apart from competing offerings? Is it feasible to develop the product?	 BUSINESS GOALS How will the product benefit the company that develops and provides it? What are the desired business benefits? Prioritise them and move the most important one to the top.
Identify your Users & Customers here: USERS - Bachelor Tenants - Micro Families - Sharing Tenants - Joint Families CUSTOMERS - Individual home Landlords - Apartment Landlords - Commercial Landlords - NRI Landlords	List the needs here: • Connecting Tenants and Landlords • Making whole rental process automated • End to end property management simplified • Reduce time for tenants to visit the properties • Simplified payment process	Name of the Product: <RentalBuddy> List 3 to 5 unique features here: → Property Listing → Price compare → Reviews → Digital Payments → Automated Documentation → Automated Services Requests	List the Business Goals here: ☐ Create continuous revenues through transaction fees ☐ Fixed fee for every agreement made ☐ Revenue through promotions

Product Vision Board Checklist:

Use the following checklist to ensure that your Product Vision Board is effective.

VISION

- ☒ **Inspiring:** Describes the positive change the product should create.
- ☒ **Shared:** Unites people, creates alignment, and facilitates collaboration.
- ☒ **Ethical:** Gives rise to a product that does not cause any harm to people and the planet.
- ☒ **Concise:** Easy to understand and remember.
- ☒ **Ambitious:** Describes a big, audacious goal that might never be fully reached.
- ☒ **Enduring:** Provides guidance for the next five to ten years and is free from assumptions about the solution.

TARGET GROUP

- ☒ **Clear:** Use relevant qualities like demographics and behavioural attributes to characterise the target group.
- ☒ **Specific:** You can tell if somebody is included in the target group or not.
- ☒ **Cohesive:** The members of a target group share similar attributes, e.g., age, lifestyle, disposable income. If that's not the case, then break up the target group and form several subgroups.

NEEDS

- ☒ **Outcome-based:** Capture the reason why people would want to use the product. Describe what success looks like from the perspectives of the users and customers.
- ☒ **Specific:** The needs are detailed enough so that you can validate them.
- ☒ **Focused:** Concentrate on the main problem/benefit, the main reason for people to use the product.
- ☒ **Prioritised:** If you do identify several needs, prioritise them according to their importance for the target group.

PRODUCT

- ☒ **Type:** It's clear what kind of product you want to offer, for example, mobile app on Android and iOS
- ☒ **Differentiated:** The aspects of your product that make it stand out, set it apart from alternative offerings are stated.
- ☒ **Focused:** There are no more than five features.
- ☒ **Big:** The features are coarse-grained product capabilities; no epics and user stories!

BUSINESS GOALS

- ☒ **Outcome-based:** The desired business benefits, the company's reason for investing in the product, are clearly described, for example, revenue, brand equity, cost savings.
- ☒ **Specific:** The business goals are detailed; state rough targets if possible.
- ☒ **Prioritised:** If more than one business goal is identified; order according to business impact.

BONUS CRITERIA

- ☒ **Needs-first:** Start with the needs after you've captured the vision especially when you create a new strategy—be it for a brand-new product or for an existing one.
- ☒ **Validated:** The strategy captured on the product vision board does not contain any major hypotheses and risks. It has been successfully validated, for instance, by interviewing and observing target users, building throwaway prototypes, and carrying out competitive analysis.
- ☒ **Adaptive:** The product vision board is regularly inspected and adapted, at least once every three months as a rule of thumb.
- ☒ **Shared:** The key stakeholders and development team members have a shared understanding of the product vision board contents and support the decisions captured on it.

Once you have created the Vision, you will do the Epic-Feature map with the help of Key stakeholders (including some users and customers) and subject matter experts. Then using some prioritization technique you will prioritize the Epic-Features and then use it as input for creating the Product Roadmap.

Epic: A big chunk of functionality which takes beyond releases (months) (Eg: User Management)

User Story: A small portion of Feature which has some business value and can be done within a Sprint (days) (Eg: A mobile user can register using his mobile number, A Facebook user can register using his/her facebook id)

```

graph TD
    EPIC[EPIC] --> FEATURE[FEATURE]
    FEATURE --> UserStory[User Story]
    UserStory --> Task[Task]
  
```

The diagram illustrates the hierarchy of product development items and their relative time scales:

- EPIC** (Orange box): Takes Months (Spans across releases on the Product Roadmap)
- FEATURE** (Blue box): Takes Weeks (Spans across Sprints, but within the Release)
- User Story** (Green box): Takes Days (Spans across Days, but within the Sprint)
- Task** (Yellow box): Takes Hours (< a Day)

Arrows indicate the flow from EPIC to FEATURE, FEATURE to User Story, and User Story to Task, showing a decreasing time scale as the items become more granular.



Sample Epic Feature Map for “RentalBuddy”

PRODUCT: RentalBuddy					
PROFILE MANAGEMENT	SEARCH	PAYMENT	DOCUMENTATION	NOTIFICATIONS	SERVICE REQUESTS
Landlord Registration	Search by Location	Configure bank details	Contract upload	Configure alerts	Raise service request
Tenant Registration	Search by GPS	Edit bank details	View contract document	Change alerts	Edit service request
Forgot password	Search by Community name	Net Banking	Edit contract document	List and view alerts	View service request
Change password	Search by Reviews	Recurring payment	Renew contract	Booking alerts	Cancel service request
Edit profile (Landlor & Tenant)	Search by type (Villa/Apartment)	Google pay	Cancel contract	Rental alerts	Raise on behalf of others
View profile details	Search by SFT range	Phone pay	View invoice details	Service request alerts	Feedback
List Profiles		Credit card payment	Print documents	Entry alerts	
Delete Profile		View bank details	Share documents	Payment alerts	
Export User Details (PDF/Excel/CSV)		Delete bank details	Download as PDF	Chat with other users	
Temporary Suspend User		List bank details		Buy/Sell alerts	
Mark user under surveillance		Adhoc payment		SOS alerts	
		List invoices			
		Partial payment			
		EMI payment			
		Refund & Adjustments			

Prioritization

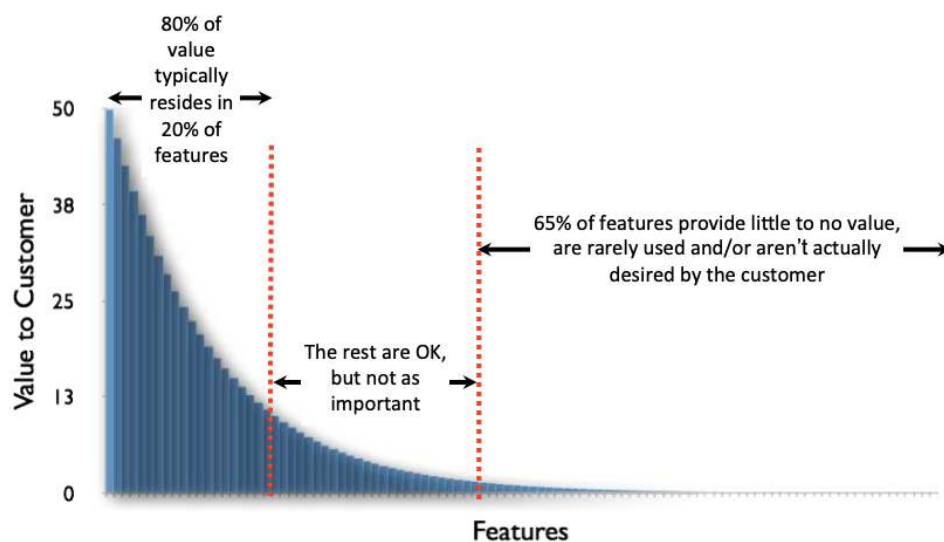
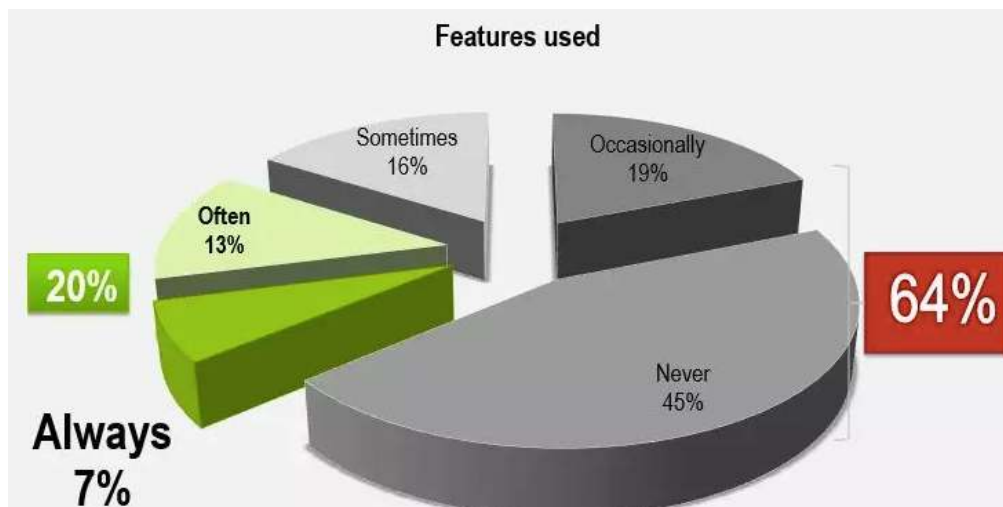
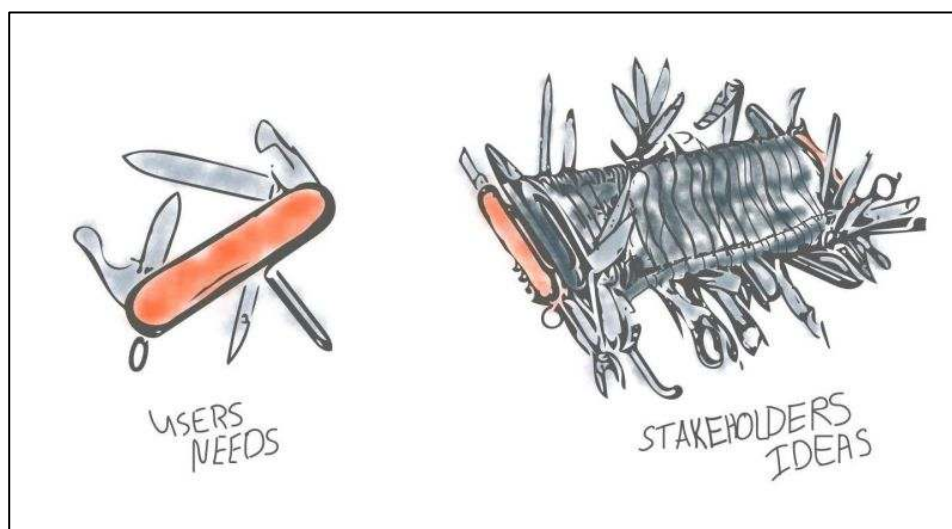


Image Source: Scrum inc.

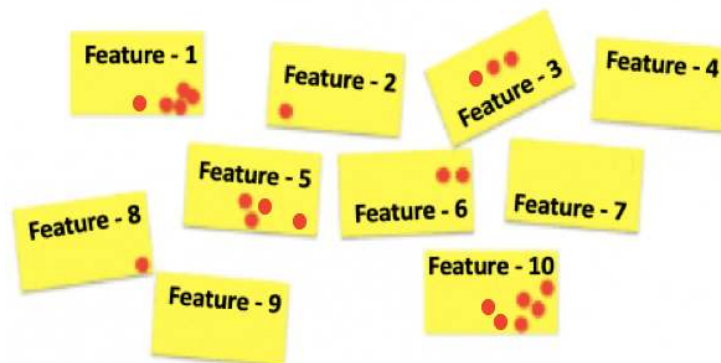


BUSINESS VALUE is the key to Prioritize the Product Backlog. Business Value can be related to different aspects as listed below:

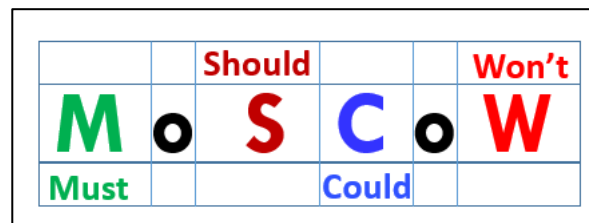
- P** Productivity Gains
- A** Avoided Costs
- R** Reduced Costs
- A** Acquire New Market
- D** Differentiation With The Competitor Products
- I** Increase Revenues
- S** Service Level Improvements
- E** Enhances Quality

Prioritization Techniques

DOT Voting



MoSCoW



MUST:

SHOULD:

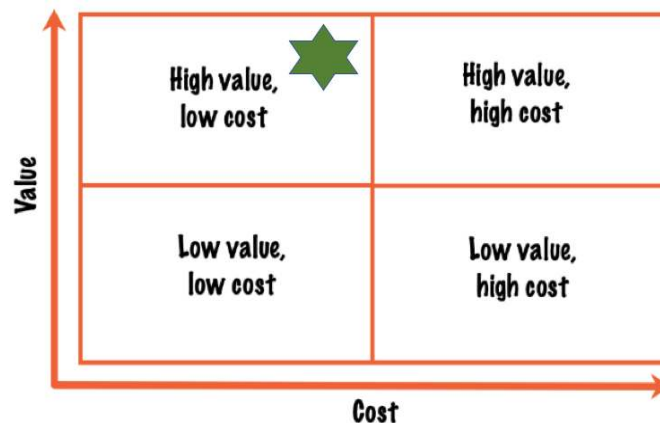
COULD:

WON'T:

Urgency Vs Importance



Value Vs Cost



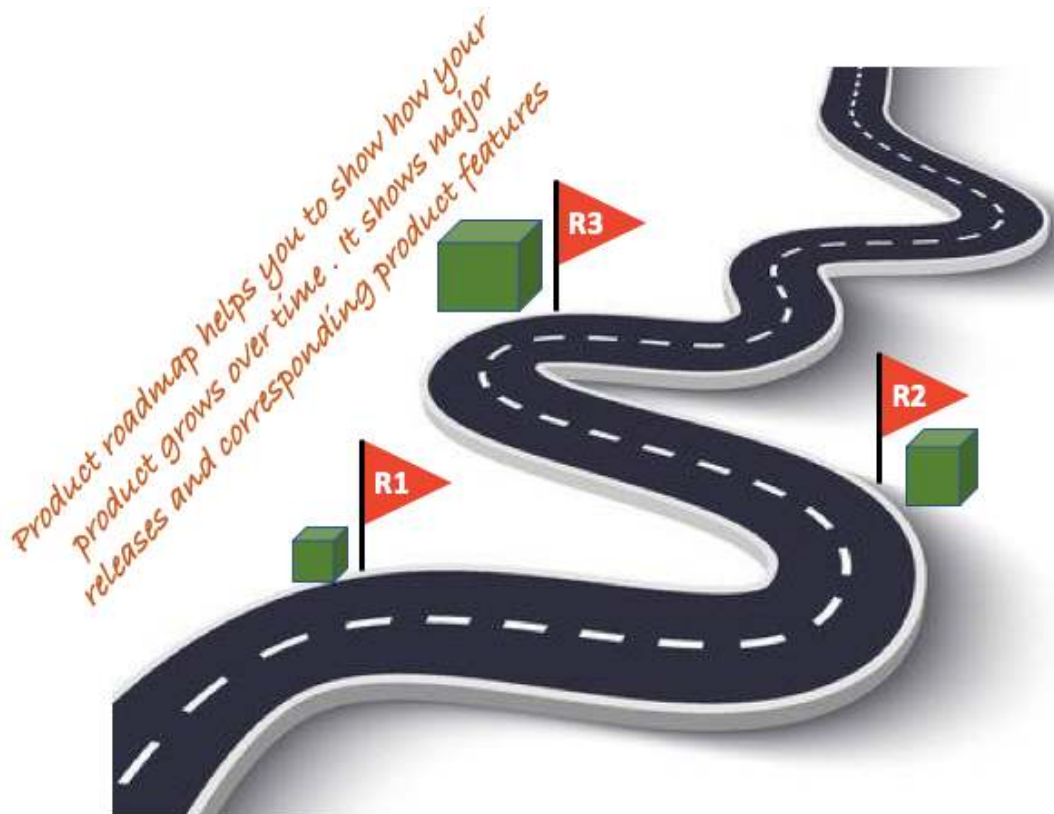
Few more methods to prioritize based on Business Value of Product Backlog Items:

- Value Poker (Relative weight by taking a reference value size and compare the rest)
- Break-even analysis (Cost of feature and expected revenue of the feature)
- Return on Investment ($[\text{Profit} / \text{Total estimated cost}] * 100$)

Epic Feature Map of “RentalBuddy” after prioritization using IMPORTANT Vs URGENT

PRODUCT: RentalBuddy					
PROFILE MANAGEMENT	SEARCH	PAYMENT	DOCUMENTATION	NOTIFICATIONS	SERVICE REQUESTS
Landlord Registration	Search by Location	Configure bank details	Contract upload	Configure alerts	Raise service request
Tenant Registration	Search by GPS	Edit bank details	View contract document	Change alerts	Edit service request
Forgot password	Search by Community name	Net Banking	Edit contract document	List and view alerts	View service request
Change password	Search by Reviews	Recurring payment	Renew contract	Booking alerts	Cancel service request
Edit profile (Landlor & Tenant)	Search by type (Villa/Apartment)	Google pay	Cancel contract	Rental alerts	Raise on behalf of others
View profile details	Search by SFT range	Phone pay	View invoice details	Service request alerts	Feedback
List Profiles		Credit card payment	Print documents	Entry alerts	
Delete Profile		View bank details	Share documents	Payment alerts	
Export User Details (PDF/Excel/CSV)		Delete bank details	Download as PDF	Chat with other users	
Temporary Suspend User		List bank details		Buy/Sell alerts	
Mark user under surveillance		Adhoc payment		SOS alerts	
		List invoices			
		Partial payment			
		EMI payment			
		Refund & Adjustments			
LEGEND OF COLORS					
IMPORTANT & URGENT					
IMPORTANT BUT NOT URGENT					
NOT IMPORTANT NOT URGENT					

Product Roadmap



Steps involved in Product Roadmap creation:

1. Gather information (Audience, Technology, Market, Feature Backlog)
2. Organize into Timespan (Monthly/quarterly/half yearly)
3. Visualize (coarse-grained Features mapped to each release)
4. Buy-in (with Stakeholders and the team)

Goal Based Roadmap

 Date The launch date or timeframe	Date or timeframe	When will the release be available?	Date or timeframe	Date or timeframe
 Name The name of the new product version or major release	Name/version	What is it called?	Name/version	Name/version
 Goal The reason for creating the new version	Goal	Why should it be developed?	Goal	Goal
 Features The high-level features necessary to meet the goal	Features	What are its key features?	Features	Features
 Metrics The metrics/KPIs to determine if the goal has been met	Metrics	How do we know the goal has been met?	Metrics	Metrics

<http://www.romanpichler.com/> 

Example Roadmap

	RELEASE 1	RELEASE 2
DATE	Q2 - 2020	Q3 - 2020
NAME	V 1.0	V 1.1
GOAL	MARKET ENTRY	ACQUIRE NEW USERS
FEATURES	EPIC: USER MANAGEMENT	
	F: REGISTER	F: LOGIN
	F: SOCIAL INTEGRATIONS	
	EPIC: SEARCH	
	F: BASIC	F: ADVANCED
	EPIC: ORDER MANAGEMENT	
	F: QUICK CREATE	F: TRACK ORDER
	F: API BASED	
	EPIC: PAYMENT MANAGEMENT	
	F: COD	F: CARDS
	F: WALLETS	
METRICS	- 100 New registrations in the first month - 25 new orders per week	- 25% increase of users month on month - 100 new orders per week

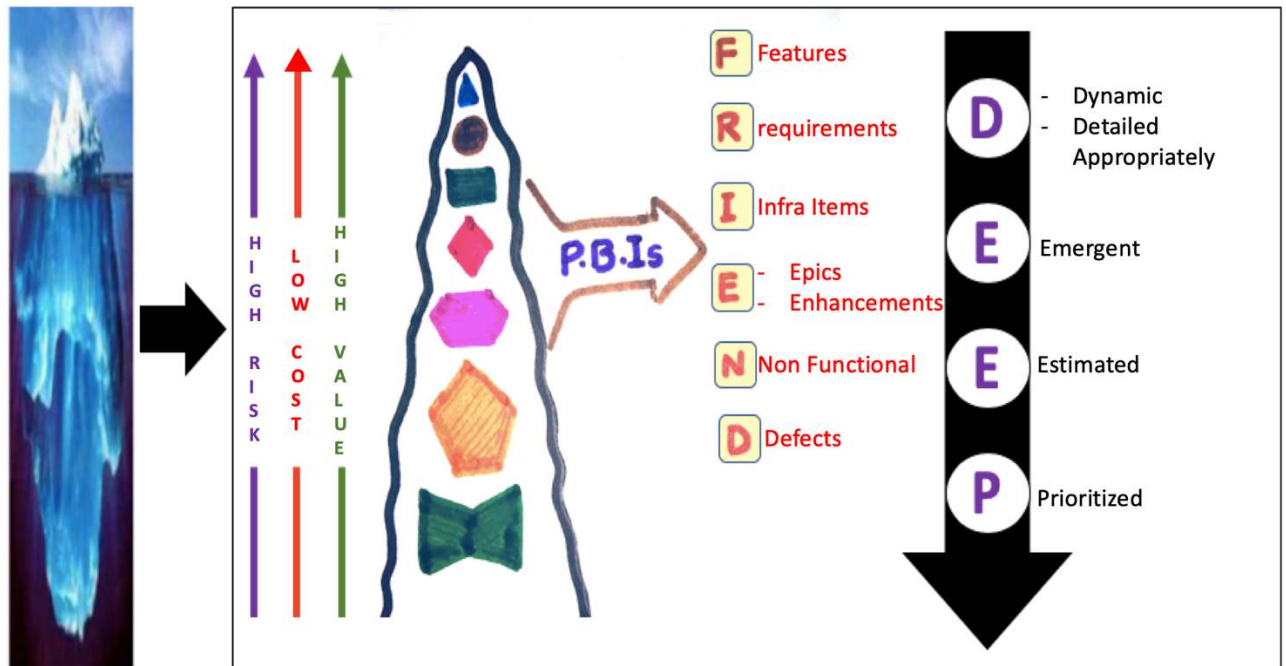
Tips to create effective Roadmaps:

1. Focus on Goals and Benefits
2. Do necessary Preparatory work
3. Tell a Coherent story
4. Keep it simple
5. Secure Strong Buy-in
6. Have the courage to say “No”
7. Know when to show dates
8. Make your Roadmap measurable
9. Determine cost top-down
10. Regularly review and adjust the Roadmaps

Mature product	Review quarterly	Review every 3 to 6 months
	Review monthly	Review quarterly
Dynamic market		Stable market

Product backlog

Product Backlog is like an iceberg,
you can see only the tip,
not the entire content !



Quick Formula for the ORDER RANK:

$$(\text{Business Value weightage} + \text{Risk weightage}) / \text{Cost weightage} = \text{ORDER RANK}$$

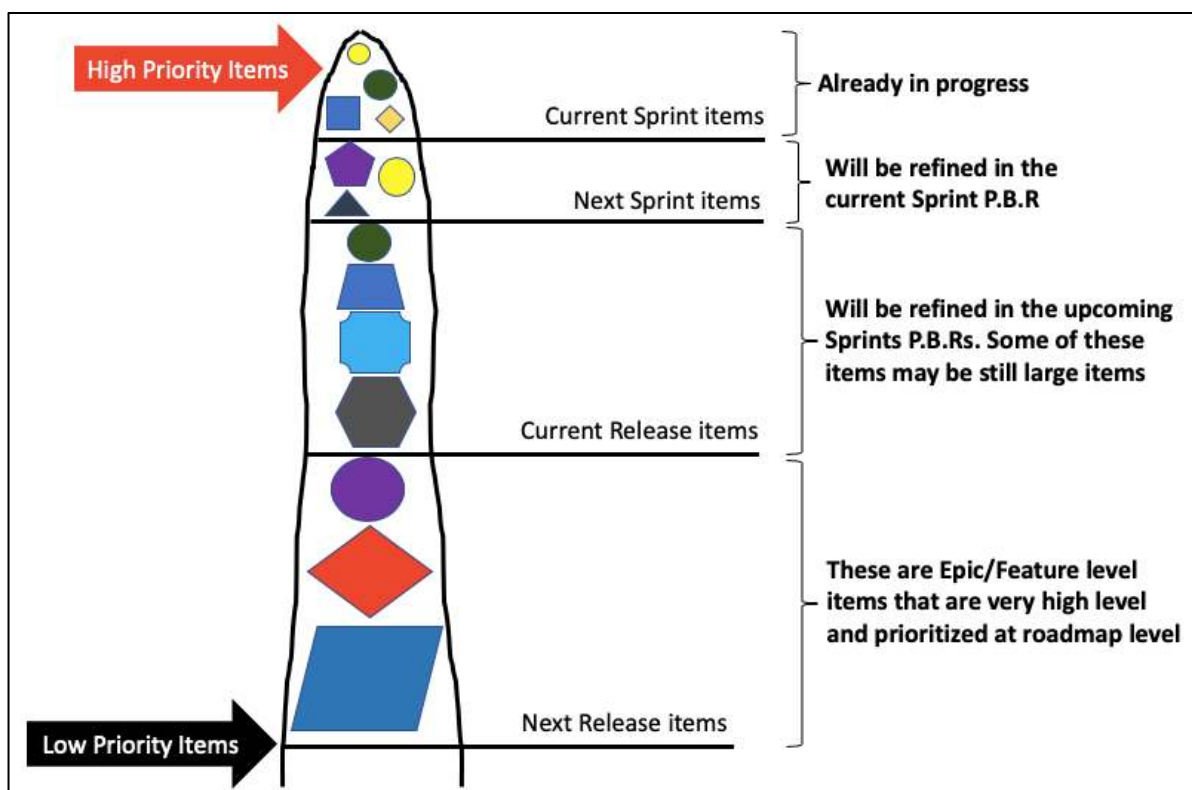
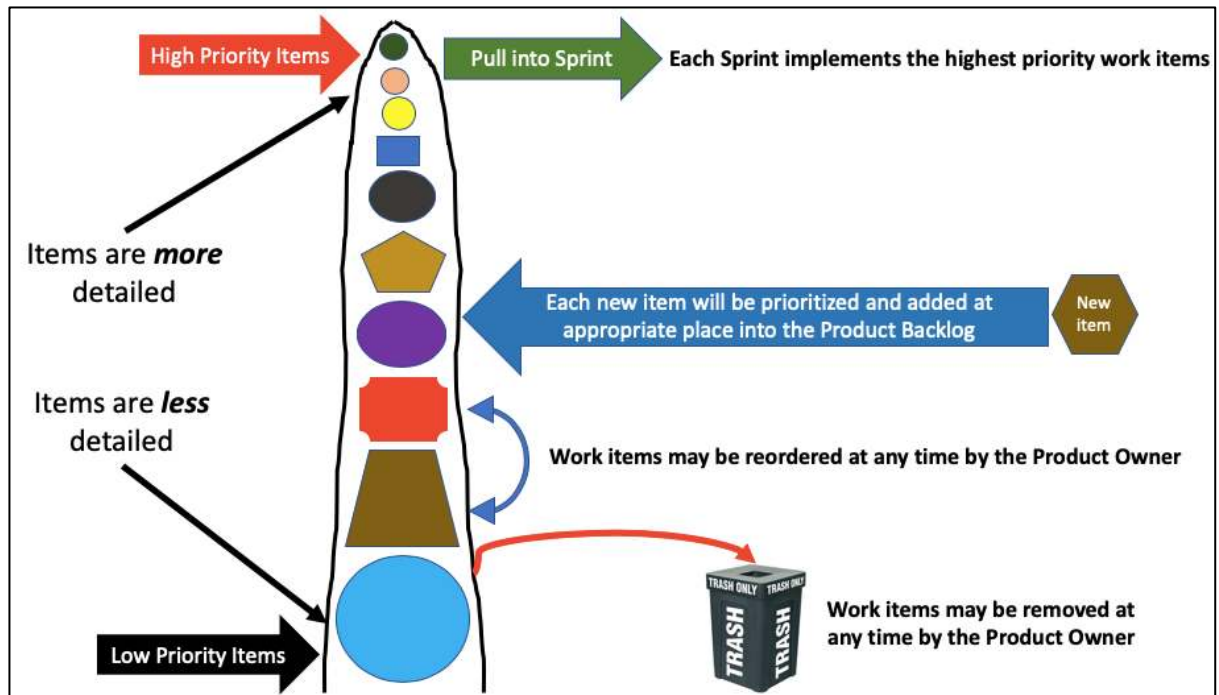
(The Highest Order Rank value items are important and on Top)

Example:

Consider 2 features of Microsoft word Save File, Print File. Give the weightage for Value, Cost and Risk for these 2 features as given below.

Save File: Value: 5, Cost: 1, Risk: 0. So as per the formula above weightage is 5 (5+0/1)

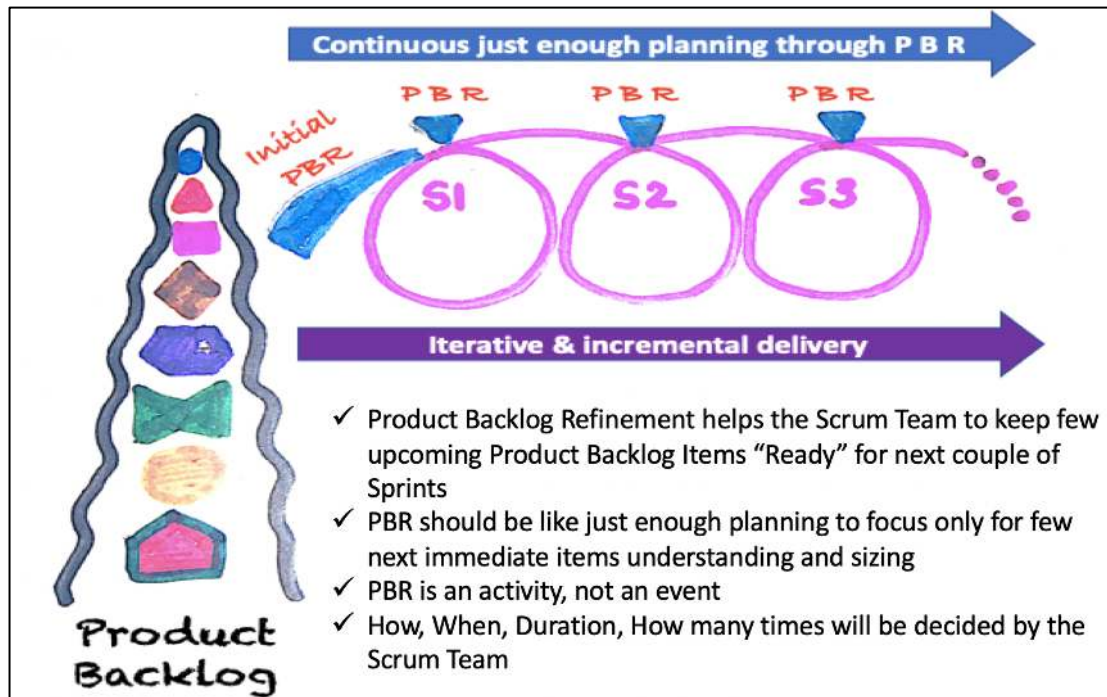
Print File: Value: 4, Cost: 2, Risk: 1. So as per the formula above weightage is 2.5 (4+1/2)



Product Backlog recap: Fill the blanks in the below story to make it complete and meaningful.

Once up on a time, there was a Product Owner named John. He started a new Product based on an idea that he has got to solve a particular problem. He created a Product **vision** to help every understand why the product is building built and what is the desired end state. John then also created a Product **Goal** that serves as a target for the Scrum Team to plan their work against. He then created a Product **Backlog** to manage all the requirements related to that product that will help to achieve the Product Goal. He always ensures that all the items that have high business **value** and important and urgent, and the items are in line with the product **Goal** are always kept on the **top** of the Product backlog. This helps John to **maximize** the product value. John allows anyone to add new items to the **Product Backlog** but he keeps the right of **prioritize** them. John takes his **Developers** help to estimate the backlog items during Product backlog **refinement** or in the **Sprint** Planning. John always keeps the Product backlog available to all so that it increases the **transparency**. John always ensures that the higher order items are **more** detailed and granular than the **lower** order items. Even though there are more teams working on the Product, John always maintains **one** Product backlog to increase **transparency** and reduce **complexity** and to **manage** dependencies across the teams effectively. The Product backlog contains the **output** (i.e., features) that helps to achieve the **outcome** (i.e., product Goal).

Product backlog refinement



What happens in Product Backlog Refinement (PBR)?

PRODUCT OWNER



- Explains the upcoming Product Backlog Items to the Development Team
- Answers the Development Team’s queries and assumptions
- Prioritizes PBIs based on Development Team’s estimates

DEVELOPERS



- Understands and upcoming Product Backlog items’ functionality
- Asks queries and clarifications to the Product Owner
- Identify technical dependencies
- Discuss high level architecture and design of upcoming items
- Works with Product Owner to split large items into smaller
- Provides estimates to the Product Backlog items

Scrum Master facilitates the Product Backlog Refinement based on the need

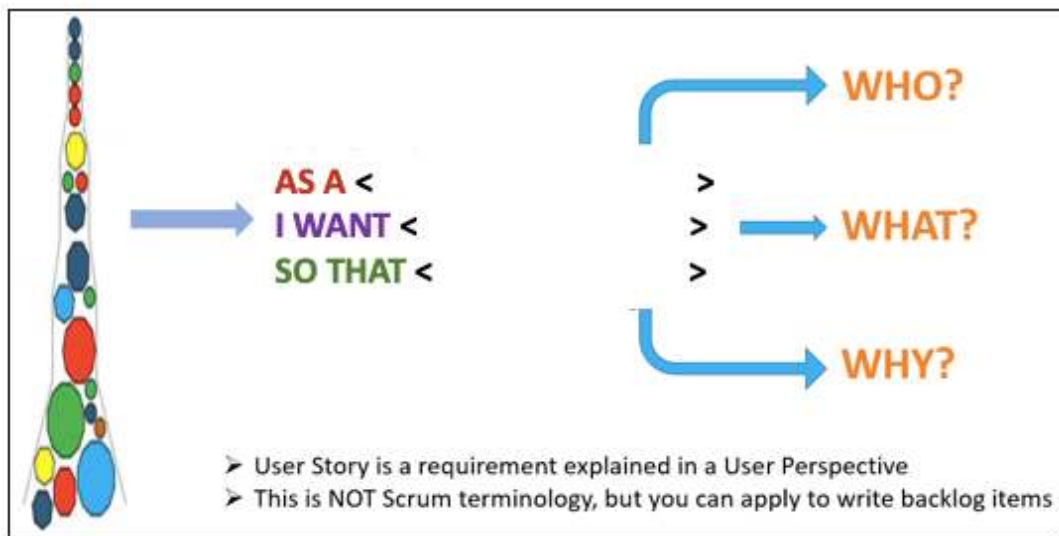


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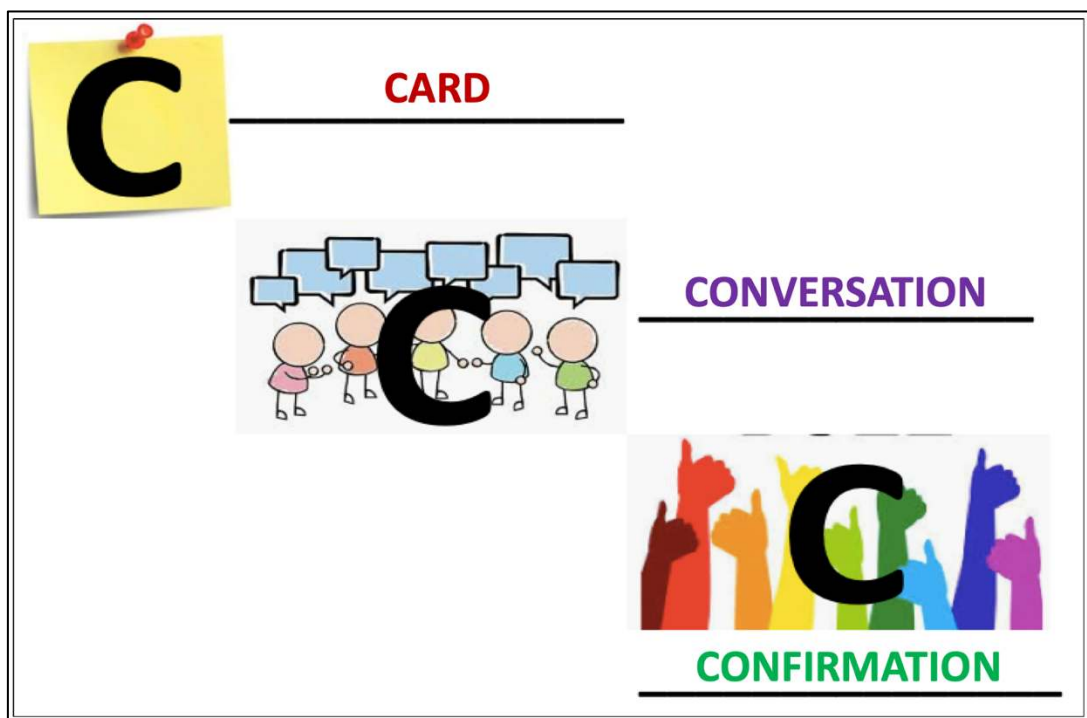
Read more about Product Backlog Refinement anti-patterns:

<https://www.learnovative.com/product-backlog-and-refinement-anti-patterns/>

User Stories



THREE “C”s of USER STORY



SPACE FOR NOTES

Sample User story with Acceptance Criteria:

Product	SUPER CRM			
User story name	Expiring proposal alerts			
User story description	AS A Sales Manager, I want to be notified by email of all expiring proposals, so that we can manage the escalations			
Release	1	Sprint	1	TESTED?
1. Notification email is sent to all sales managers on 15th and 30th of every month contains a summary of all expiring proposals (Summary email template is attached with placeholders)			YES	YES
2. If there is a leap year, then for February month the email should be sent on the 28th Feb			YES	YES
3. Summary email template has some clickable links which should take the user to expiring proposals			YES	
4. Subject of the email should be "Proposals expiring before DDth of this month" (dd: 15 or 30)				

Another example user story:

User Story: As a Visa credit card user, I should be able to pay for the items in the shopping cart using his/her card so that I can complete my shopping seamlessly

Acceptance criteria:

1. User credit card should be a Visa card
2. All other cards like Master, Amex cards payment should not be accepted
 - a. Message to show "Only Visa cards are allowed"
3. User card should have valid number of digits
4. Card CVV must be valid and 3 digits length
 - a. If invalid CVV then: "Wrong CVV, please try again"
5. Cards with expired should be rejected
 - a. Message: "Your card has expired, please try with another card"
6. Card holder name must be validated
 - a. Message: "Invalid card, please provide correct card holder name"
7. Different purchase amounts should work with positive amounts
8. Zero or negative amounts should be alerted
 - a. Message: "Amount should not be negative number"
9. Up on successful payment, user should receive a confirmation on:
 - a. Email with the payment transaction number and date
 - b. Text message with the payment transaction number and date

Good acceptance criteria should be: **SAFE**

Success: What is the success criteria?

Example: If “Login” is the user story then after successful logging on, where the user should land. May be edit profile, or search or my dashboard etc.

Advance: What has to happen before the success outcome can be reached?

Example: From the user story begin till success there may be steps like accepting the terms and conditions checkbox, or enter the OTP in order to move advance towards the success of the user story.

Failure: What could go wrong and how can user be coped up?

Example: Accidentally entered wrong PIN number for an ATM transaction which can be addressed with a message and a retry in same transaction. Failure is something that can be recovered.

Error: What are the error situations and what messages to be shown?

Example: When user tries to withdraw money from an ATM machine but the transaction has some server connection issue, error is something that cannot be recovered.



SPACE FOR NOTES

“INVEST” of User Stories

Every user story should satisfy the “INVEST” principle in order to be a good user story. This helps Scrum Teams to make it into working software quick and can deliver value to the customer.

Independent: Stories should be as independent as possible. When thinking of independence it is often easier to think of “order independent.” In other words, stories can be worked on in any order. Why is this important? It allows for true prioritization of each and every story. When dependencies come into play it may not be possible to implement a valuable story without implementing other much less valuable stories. Also, the more dependency between the Stories, the more delays in value delivery and it takes additional (waste) effort of regression testing if a user story gets a change in future the dependent stories also to be tested.

Negotiable: A story is not a contract. A story is an invitation to a conversation. The story captures the essence of what is desired. The actual result needs to be the result of collaborative negotiation between the Product Owner, Developers. The goal is to meet customer needs, not develop something to the letter of the user story if doing so is insufficient!

Valuable: If a story does not have discernible value it should not be done. Period. Hopefully user stories are being prioritized in the backlog according to business value, so this should be obvious. Some people say each story should be valuable to the customer or user. I don’t like that way of thinking because business value encompasses more than just customer or user facing value. It includes internal value which is useful for things which are normally called “non-functional requirements” or something similar. I prefer to say the story has value to the “user” in the user story. In this way it is clear who is to be satisfied. Finally, remember the “so that <value>” clause of the user story. It is there for a reason – it is the exact value we are trying to deliver by completing the story!

Estimable: A story has to be able to be estimated or sized so it can be properly prioritized. A value with high value but extremely lengthy development time may not be the highest priority item because of the length of time to develop it. What happens if a story can’t be estimated? You can split the story and perhaps gain more clarity. Sometimes splitting a story doesn’t help though. If this situation occurs it may be necessary to do some research about the story first. Please, please, please timebox the research! If you do not, it will take all available time thereby depriving the product of something else which could have been done instead.

Small: Obviously stories are small chunks of work, but how small should they be? The answer depends on the team and the methodology being used. I teach agile and suggest two week iterations which allow for user stories to average 3-4 days of work – TOTAL! This includes all work to get the story to a “done” state. Also remember not to gold plate user stories. You should do the simplest thing that works – then stop!

Check out our extensive resources on how to split user stories.

Testable: Every story needs to be testable in order to be “done.” In fact, I like to think of testable meaning acceptance criteria can be written immediately. Thinking this way encourages more collaboration up front, builds quality in by moving QA up in the process, and allows for easy transformation to an acceptance test-driven development (ATDD) process. As with negotiable above, asking the magic question can help ensure the user story is testable as well.

If Product Owners and their teams work together to INVEST in good user stories the learning curve of working together will be much shorter. INVEST encourages good habits which eliminate some of the bigger problems of user stories like dependencies, being too big, hard to test, etc. Take the time to INVEST in good stories and see the dramatic change in how effective planning will become, as well as how productive the team will become.

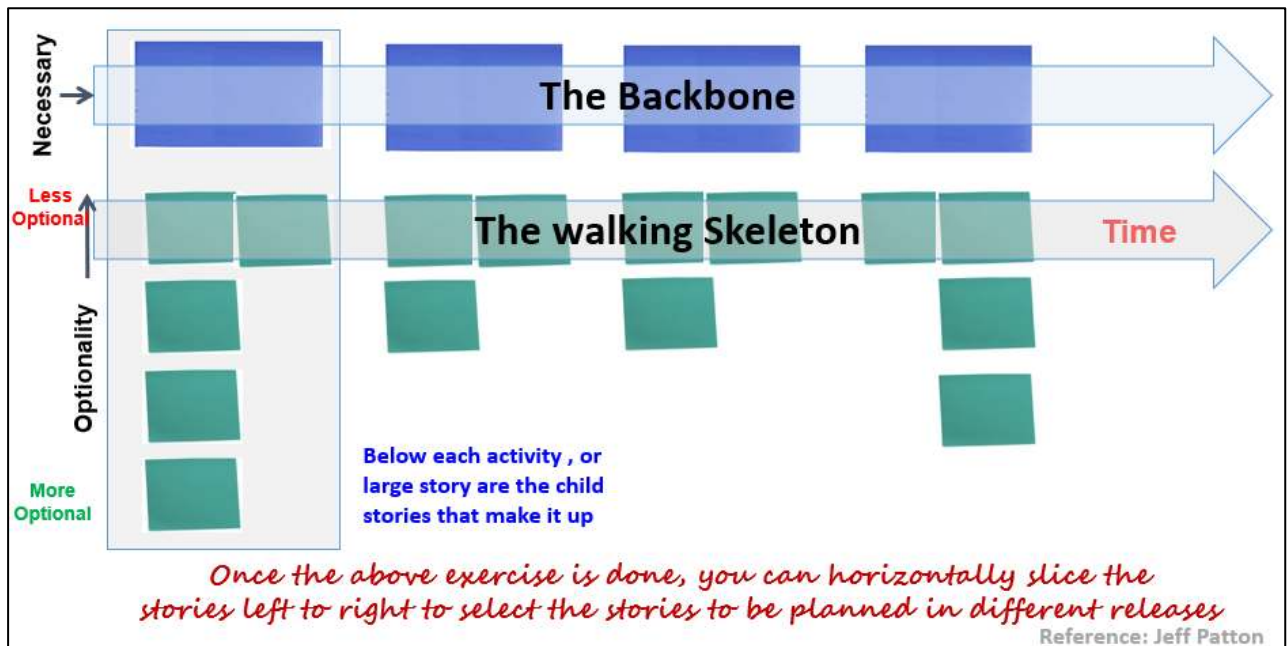
Guidelines for writing Acceptance Criteria:

Acceptance criteria aren't easy to write. Despite the simple format, writing the text is a challenge. Here are seven tips to help you avoid common mistakes while writing acceptance criteria or to review criteria written by a member of your team.

- **Document criteria before the development process starts.** In this way, the team is more likely to catch all of the client's needs in advance. Initially, it's enough to set criteria for a small number of user stories to complete backlogs for two sprints. The documented acceptance criteria are then used by developers to plan the technical process.
- **Don't make acceptance criteria too narrow.** Acceptance criteria that's too specific leaves developers little room to manoeuvre. To avoid this, remember that acceptance criteria should be an expression of intent, not a final decision. Moreover, narrow acceptance criteria may not take into account all user actions.
- **Keep your criteria achievable.** Effective acceptance criteria define a reasonable minimum amount of functionality that you can provide. But if you keep describing all the little details, there's a risk that your team will get stuck on hundreds of small tasks.
- **Avoid too broad of acceptance criteria.** Broad acceptance criteria make a user story vague. Effective acceptance criteria must outline the scope of work so that developers can properly plan and estimate their efforts.
- **Avoid technical details.** Acceptance criteria should be written in plain language. Your stakeholders and managers may not have technical backgrounds, so using plain language will make the criteria understandable for everyone.

- **Reach consensus.** The same problem can be solved in different ways by team members and stakeholders depending on their points of view. Make sure you communicate your acceptance criteria to stakeholders and team members and reach a mutual agreement.
- **Write testable acceptance criteria.** This will give testers the opportunity to ensure that all requirements are met and will allow developers to know if the user story is complete.

Story mapping



User Story Mapping is an approach to Organizing and Prioritizing user stories. A powerful way to gain an understanding of scope for new product development. These are typically told from the end-user perspective. Unlike typical user story backlog, Story maps:

- ✓ Make visible the work flow or value chain
- ✓ Show the relationships of the larger stories to their child stories
- ✓ Help confirm the completeness of your backlog
- ✓ Provide a useful context for prioritization
- ✓ Plan releases in complete and valuable slices of functionality

The foundational building block of a story map is the "User Task"

How do we do this?

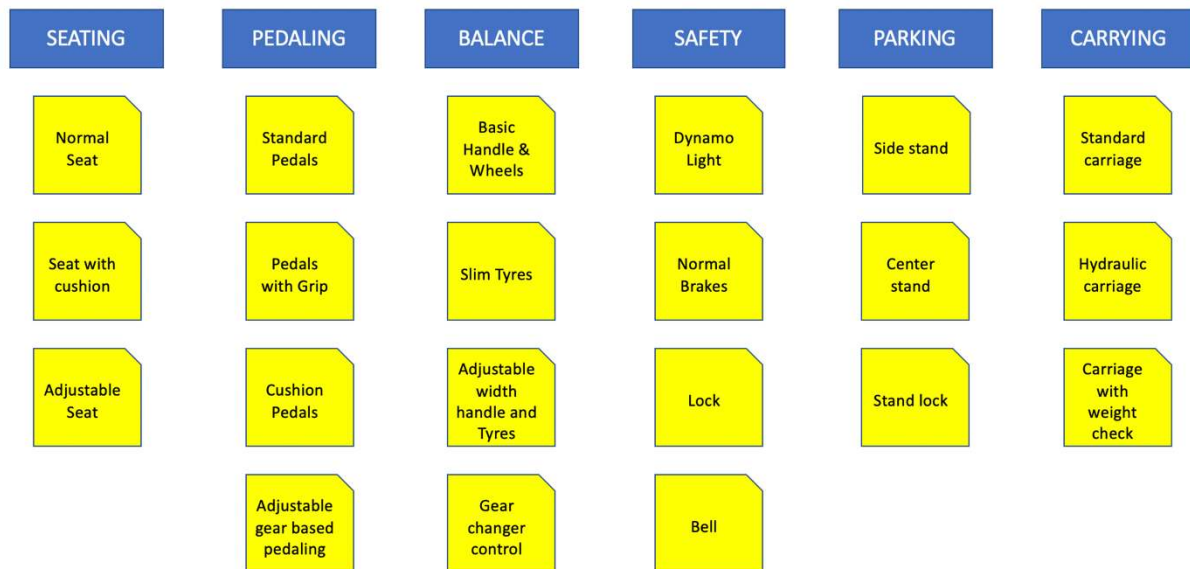
- Start with an understanding of user experience and define user roles
- Write user scenarios to think through user experience
- Imagine the user experience as a scenario
- Extract task-centric user stories from your user experience
- Organize user stories into a map that communicates experience



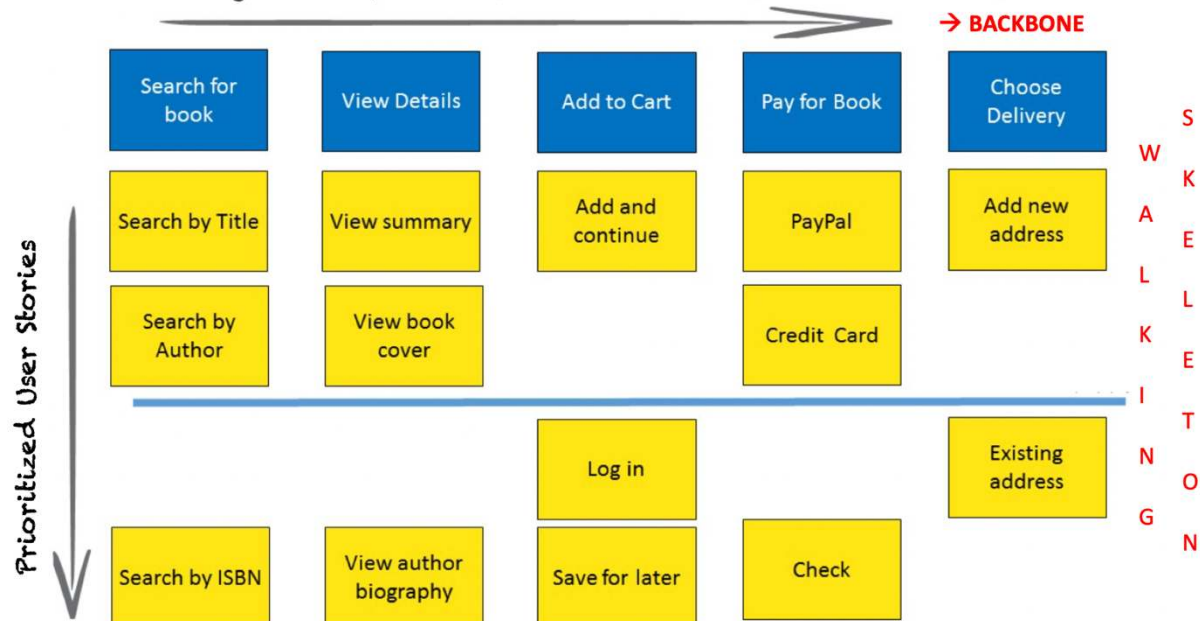
SPACE FOR NOTES

CERTIFIED SCRUM PRODUCT OWNER (CSPO) WORKSHOP

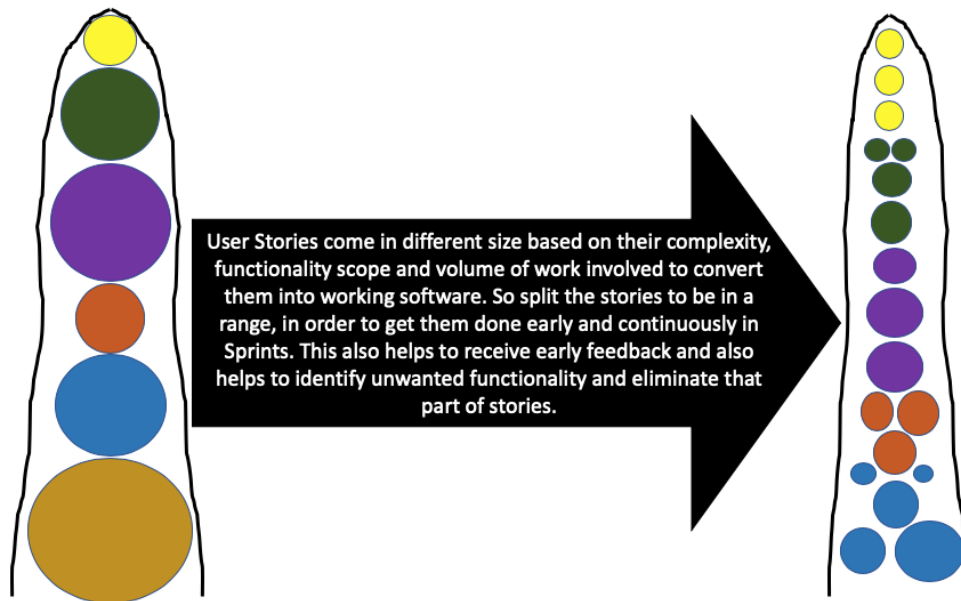
Examples for Story map:



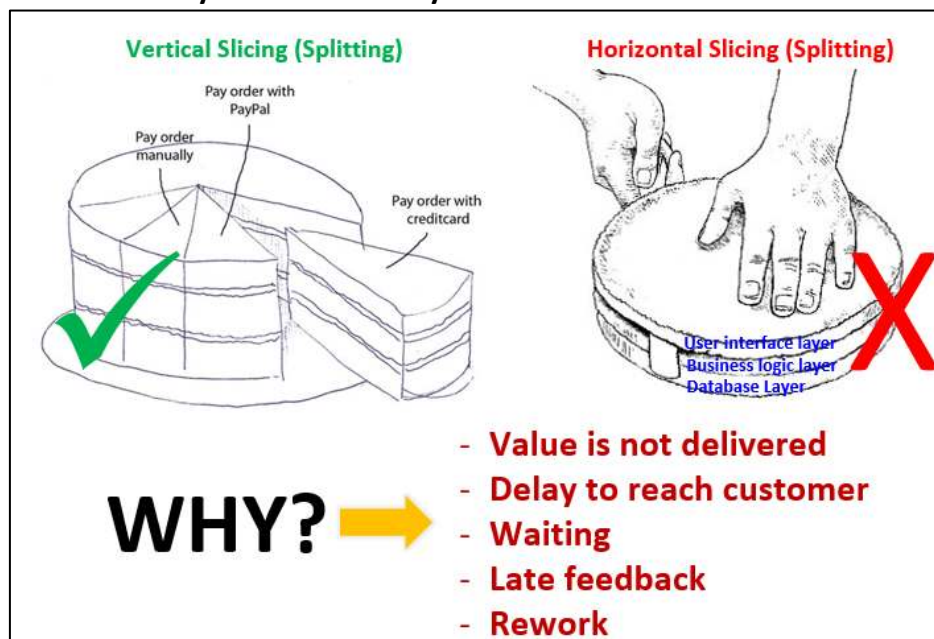
Activity - The steps or set of tasks that work together to achieve a goal



Splitting the user stories



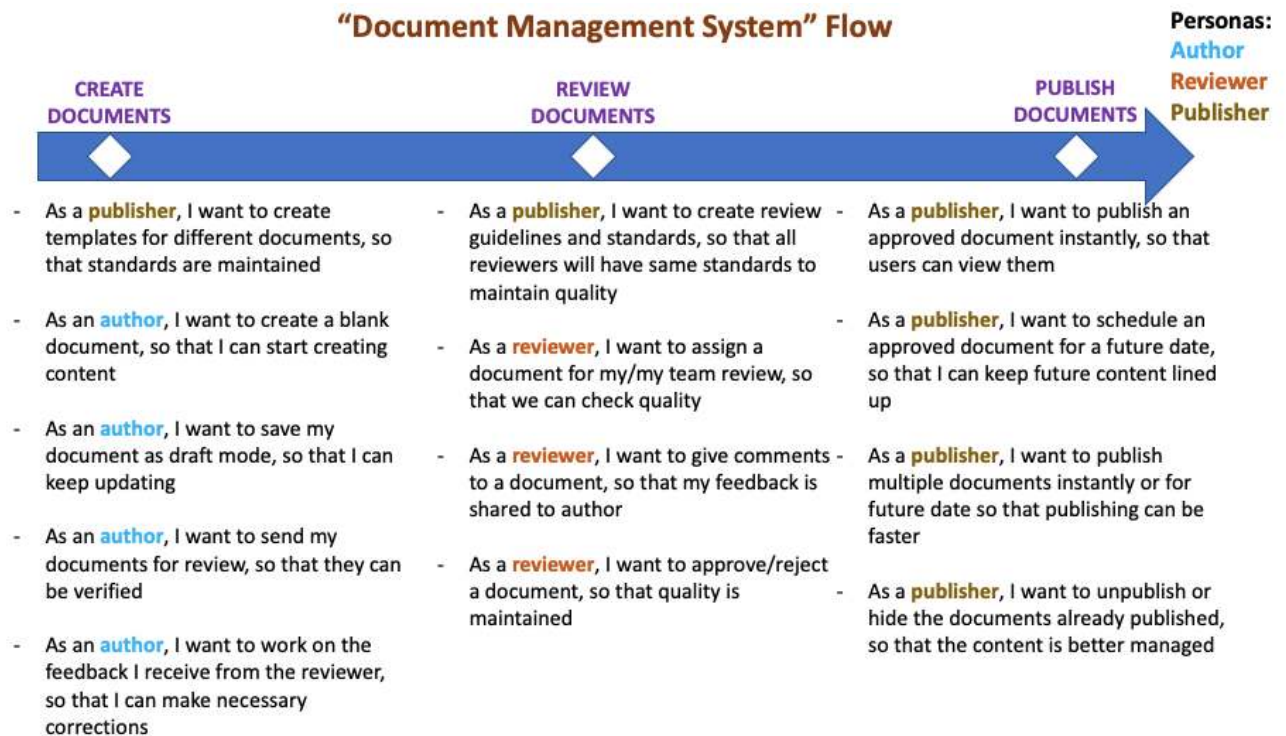
Split the Stories vertically NOT horizontally



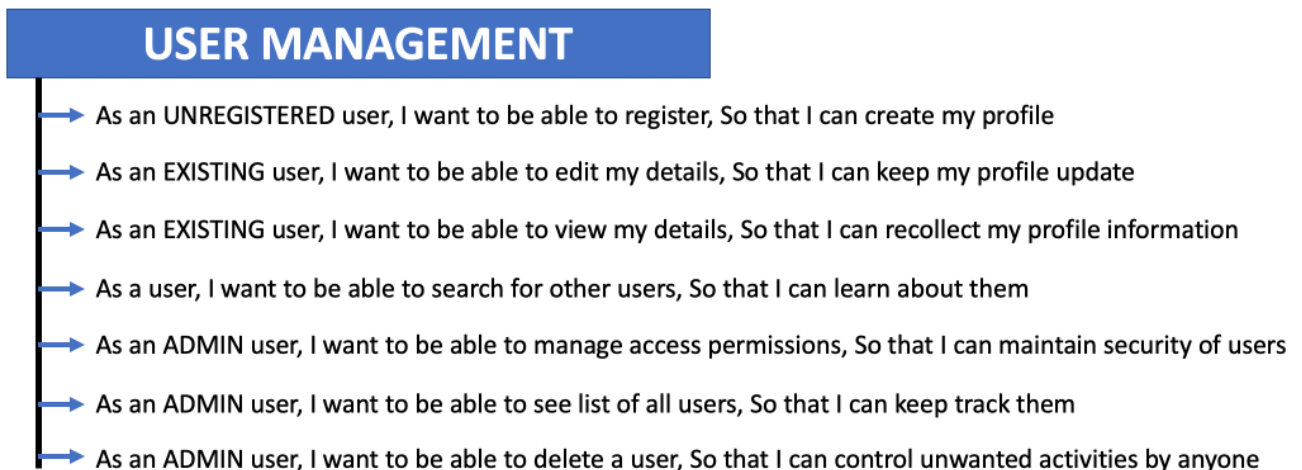
Splitting Techniques:

- Work flow steps pattern
- Operations pattern (CRUD)
- Simple to complex pattern
- Data entry patterns
- User Role/Persona based pattern
- Mock UI pattern
- Defer performance pattern
- Breakout a Spike pattern

Example for Workflow steps pattern splitting:



Example for CRUD operations pattern splitting:



Simple to complex pattern splitting:

Book a Flight

☒ One Way
 ☐ Round Trip
 ☐ Multi City

From
Select City

To
Select City

Departure Date

Return Date

Passenger(s)
1

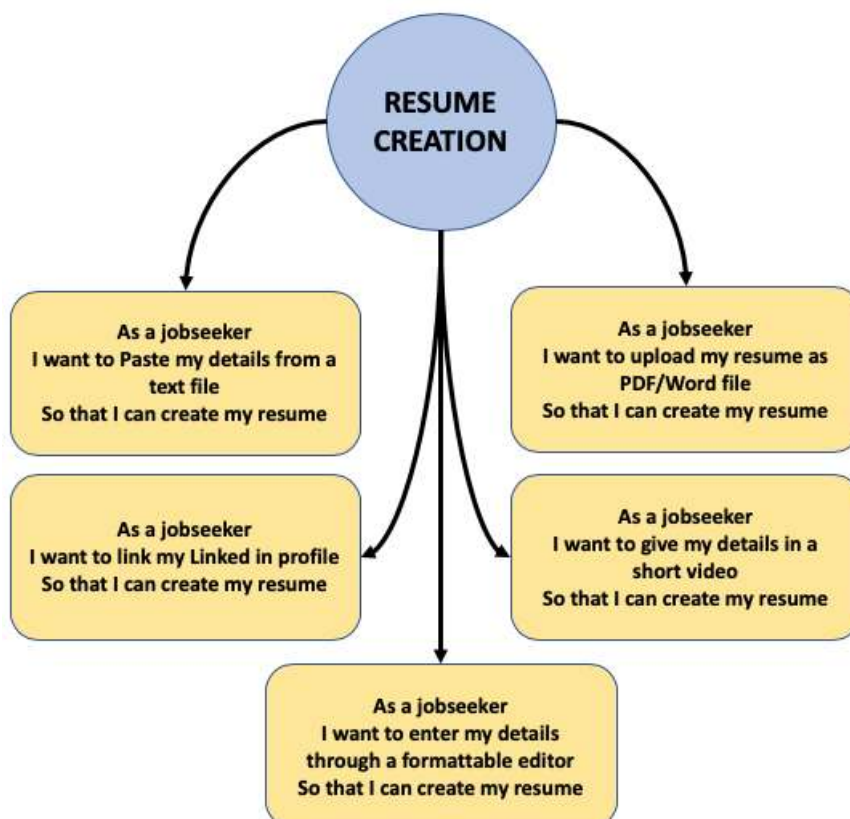
Pay in Currency
Indian Rupees

☐ Armed Forces
 ☐ Senior Citizen
 ☐ Family & Friends
 ☐ Students
 ☐ Minor

Search Flight →

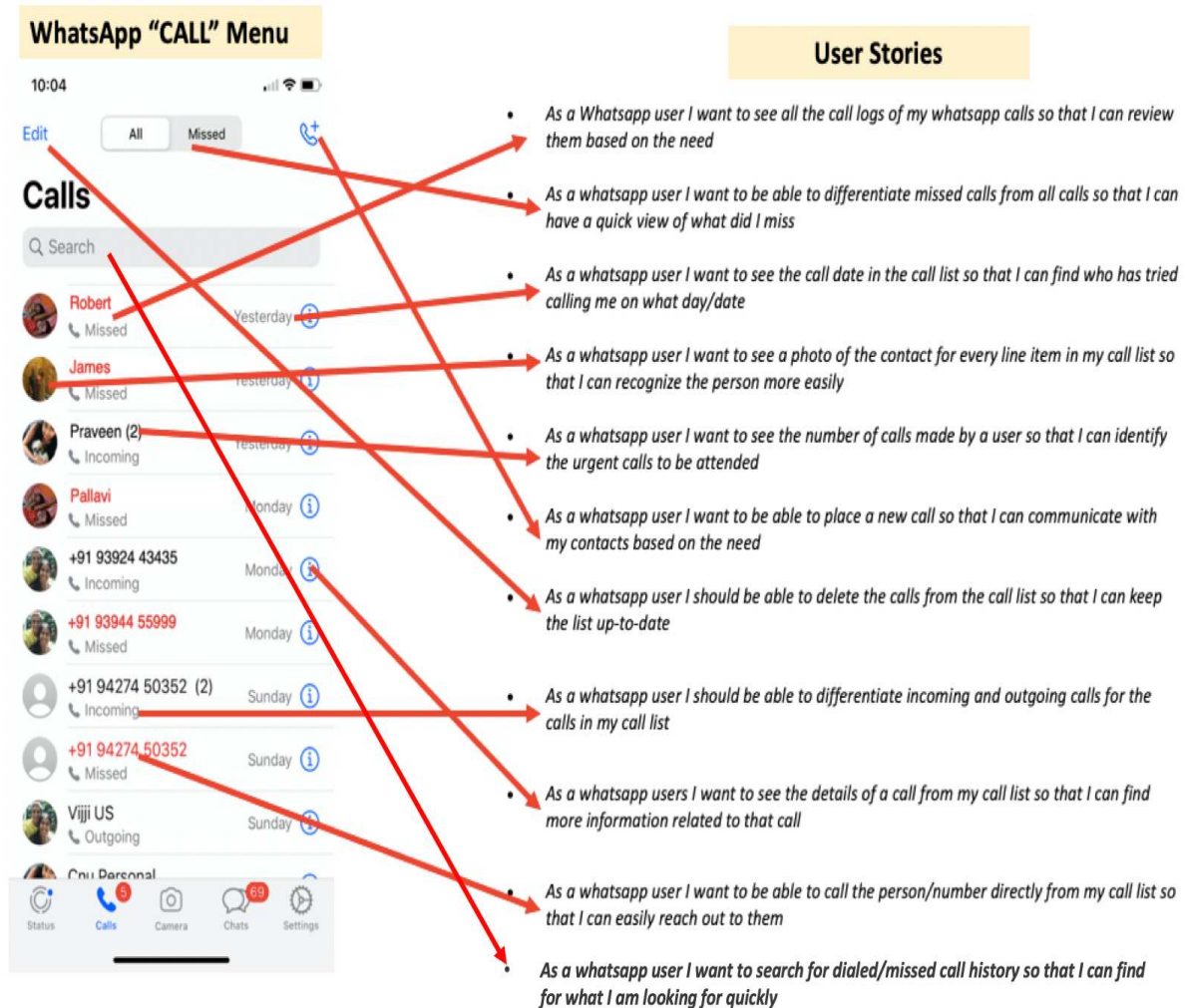
- As a **traveler**, I want to search for one way travel for a city to another city for a given date, so that I can plan my travel
- As a **traveler**, I want to search for round-trip travel for a city to another city for a given date, so that I can plan my travel with confirmed return ticket
- As a **travel agent**, I want to search for multi-city travel for a route between two dates, so that I can plan my customer's vacation
- As a **traveler**, I want to choose my payment currency type for my booking, so that I can have flexibility in payment
- As a **Senior Citizen**, I want to select my travel date from a calendar, so that I can avoid wrong bookings that may happen by manual typing of dates
- As a **traveler**, I can select different age group members for my travel, so that I can manage my family holiday plan

Data entry pattern splitting:

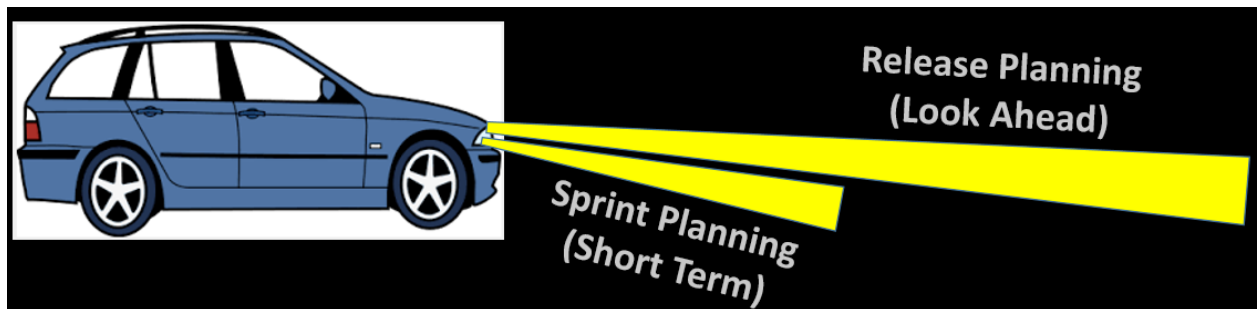


Sample User Stories

Below is the WhatsApp Call log feature and corresponding user stories.



Release Planning (Not part of core Scrum)



A very high-level plan for multiple sprints (E.g. 4 – 6 Sprints) is done during release planning. It is a guideline that reflects expectations about which features will be implemented and when they are completed. It also serves as base to monitor progress with the project.

Releases can be intermediate deliverables done during the project or the final delivery at the end.

Prerequisites to create a Release Plan:

- A prioritized and estimated Product Backlog
- The estimated Velocity of the scrum team
- Conditions of satisfaction (goals for the schedule, scope, resources, cost etc.)

Depending on the type of project (Feature driven or Date driven) the release plan can be created in different ways:



DATE DRIVEN
(Time Based Release)



FEATURE DRIVEN
(Scope Based Release)

FIXED SCOPE (FEATURE DRIVEN) RELEASE PLANNING:

STEP 1:

The first step in fixed-scope release planning is to groom the product backlog to include at least the product backlog items (PBIs) the organization must have in this release by creating, estimating the size of, and prioritizing the PBIs that comprise the Minimally Releasable Features.

STEP 2:

The second step in fixed-scope release planning is to determine the total size of the PBIs to be delivered during the release. Do this by summing up the estimates of the must-have PBIs.

STEP 3:

The third step in fixed-scope release planning is to measure or estimate the team's velocity as a range.

STEP 4:

The fourth step in fixed-scope release planning is to divide the total size of the PBIs for the release by the **fastest** velocity in the range. Round the answer up to the next integer. This will indicate the lowest number of sprints required to deliver the features targeted for the release.

STEP 5:

The fifth step in fixed-scope release planning is to divide the total size of the PBIs for the release by the **slowest** velocity in the range. Round the answer up to the next integer. This will indicate the greatest probable number of sprints required to deliver the features targeted for the release.

EXAMPLE

When asked how many sprints it will take to deliver the fixed scope, you can now answer using a range: somewhere between the greatest probable number of sprints and the least number of sprints. Suppose, for example, your features targeted for a release is estimated to be 150 points, and the Scrum team's predicted velocity range is 18-22 points per sprint. You can estimate that the MRF will be delivered in 7-9 sprints. If you are running 2-week sprints, this would equate to 14-18 weeks.

FIXED DATE (DATE DRIVEN) RELEASE PLANNING:

STEP 1:

The first step in fixed-date release planning is to determine how many sprints are in the release. If all sprint lengths are equal (which should be the ideal case), this is simple calendar math, because you know when the first sprint will start and you know the delivery date.

STEP 2:

The second step in fixed-date release planning is to groom the product backlog to a sufficient depth by creating, estimating the size of, and prioritizing product backlog items. You'll need enough PBIs to plan out to the fixed release date.

STEP 3:

The third step in fixed-date release planning is to measure or estimate the team's velocity as a range.

STEP 4:

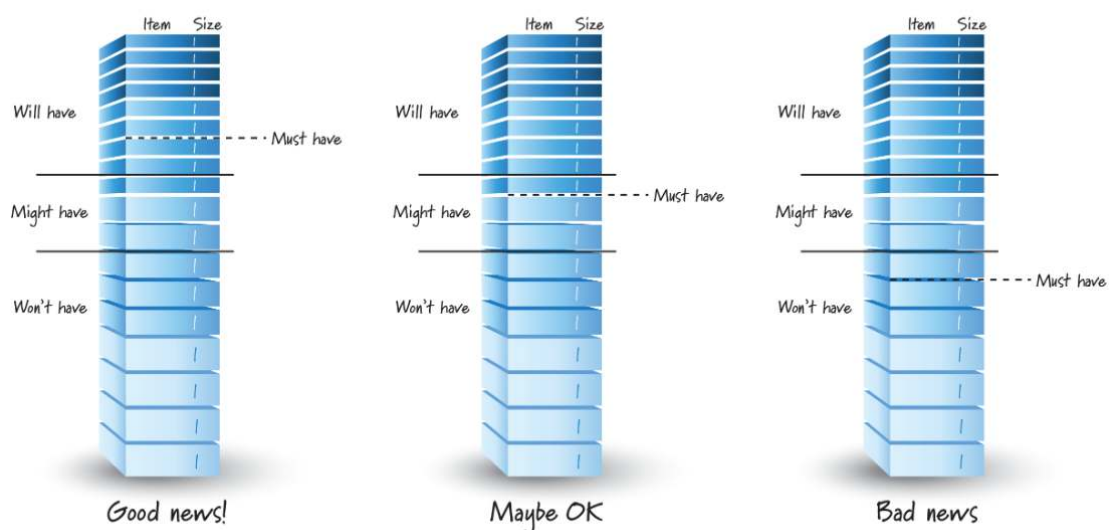
The fourth step in fixed-date release planning is to create a "will have" line by multiplying the slowest number in the velocity range by the number of sprints. Count down that number of points into the product backlog and draw a line.

STEP 5:

The fifth step in fixed-date release planning is to create the "might have" line by multiplying the largest number in the velocity range by the number of sprints. Count down that number of points into the product backlog and draw another line.

STEP 6:

The final step in fixed-date release planning is to compare the "will have" and "might have" lines to the minimum releasable features (MRFs) in the product backlog. If all of the must-have items are above the will have line, you are in great shape. If not, you will need to decide whether to proceed as planned, revisit the MRFs, pivot, or terminate the project. The image below shows several variations in how the must-have features (MRFs) might compare to the will have and might have calculations from fixed-date release planning.

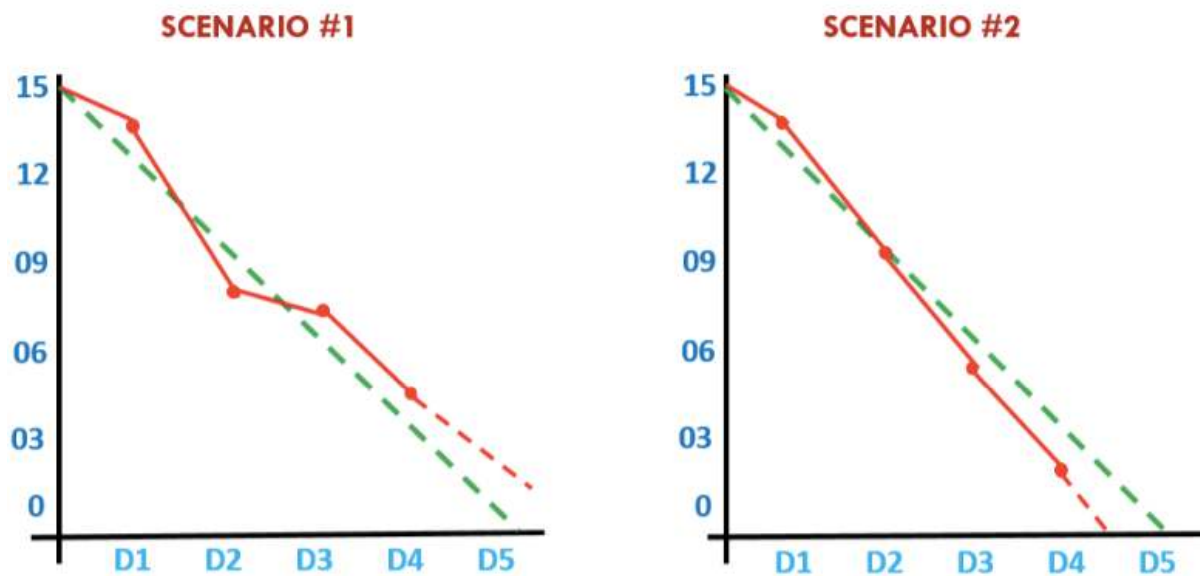


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After the Release Planning is done, the Scrum team starts Sprinting to create potentially releasable product increment. Product Owner closely collaborates with the Developers and the Stakeholders to make effective decisions based on collaborative discussions with them.

During each Sprint, the Developers track the progress of the work of Sprint using some tracking mechanism. Most common approach is a “Sprint Burndown”.

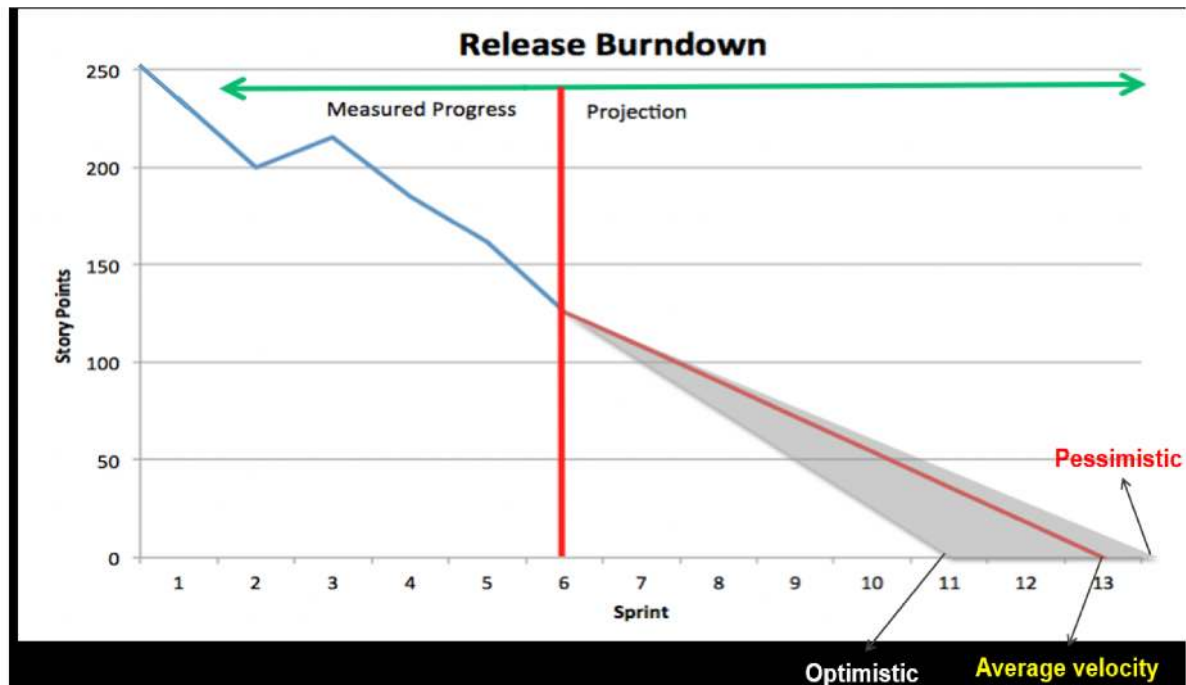
Below are some example Burndown charts of a team which has different scenarios:



The Developers update the Sprint Burndown chart. Ideally it should be updated as and when a user story is “Done”, or else at least once in a day it should be updated.

Tracking Progress through release Burndown chart:

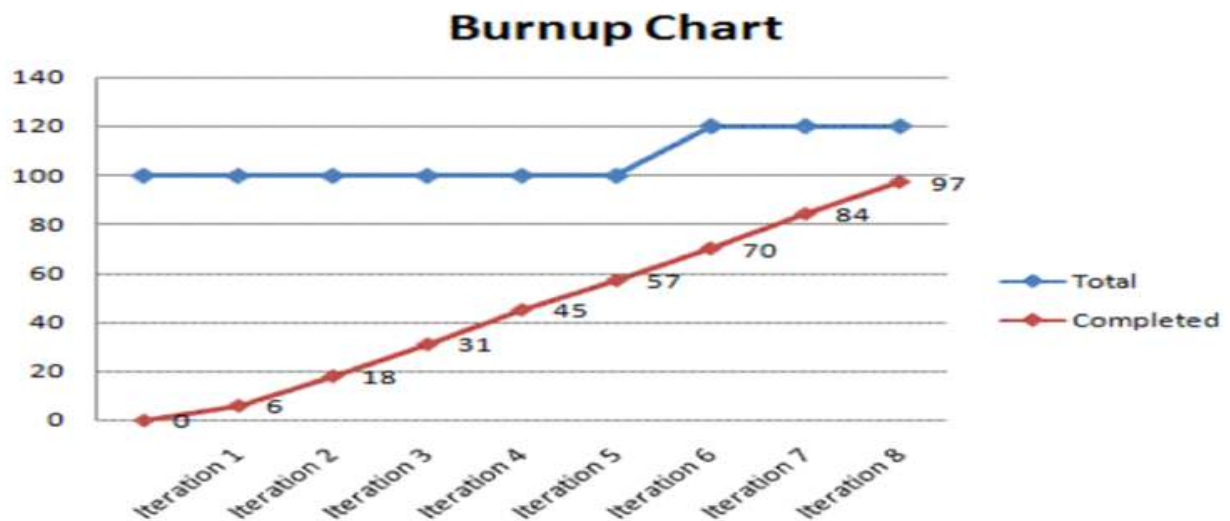
Product Owner updates the Release Burnup/Burndown chart at the end of every Sprint. This helps the Product Owner to update Stakeholders with the progress and also to make scope vs time trade-off decisions.



Total story points at the beginning were 250. As per initial release planning, they estimated to take 13 Sprints to complete the 250 Story Points worth of backlog items. Currently the status is, 6th sprint completed and the remaining story points as of now were 125 and remaining Sprints are 7 as per the initial release plan. Based on the historical velocity (by taking the individual velocity of the 6 Sprints completed) the Product Owner can make below forecast:

- ✓ Release is 2 Sprints ahead of schedule considering the best velocity
- ✓ Release is 0.5 Sprints behind schedule considering the worst velocity
- ✓ Release is on track considering the average velocity

Tracking Progress through release Burnup chart:



Blue line (Top) indicates the overall release backlog size. Till the 5th Sprint, it is same but in Sprint 6 it has increased, that means some new work is added. Red line (bottom) indicates the cumulative completion of the work in every Sprint. So, at the 8th Sprint, “Remaining work to be complete” can be found by subtracting red line value (work complete) from the blue line value (Total work). So, in the above it is 120 minus 97 which is 23. Now, to understand how many more sprints it is needed to complete the 23 points of work, you have to check the average velocity of all 8 Sprints. In the above it is: 12.125, let’s take 12. So now divide the 23 (remaining work) with 12 (Velocity) which gives the number of Sprints required to complete. $23/12 = 2$ (rounded). This is how the Product owner will track the overall release progress.

Velocity

Velocity is a measure of the amount of work a Team can tackle during a single **Sprint**.

Velocity is calculated at the end of the **Sprint** by totalling the Story Points for all **fully completed** backlog items.

- *Velocity is a Forecast, not Commitment*
- *Stories that are 100% done only will be considered to calculate velocity*
- *You cannot compare teams based on Velocity*
- *It is at the team level, not at Individual level*



SPACE FOR NOTES

APPENDIX

Space for Notes

Space for Notes

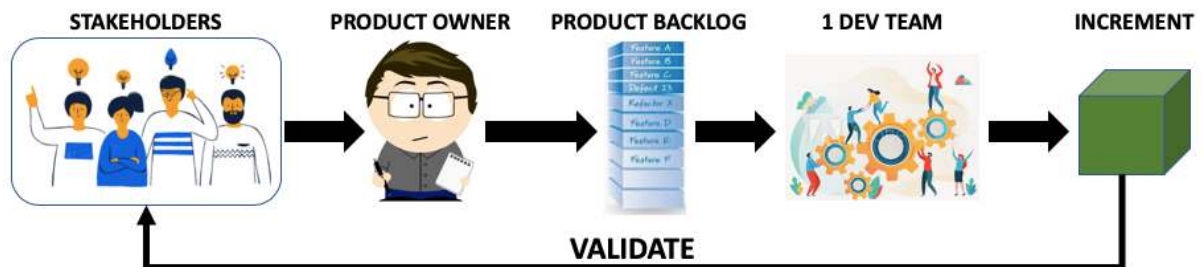
Space for Notes

What makes you a Great Product Owner?

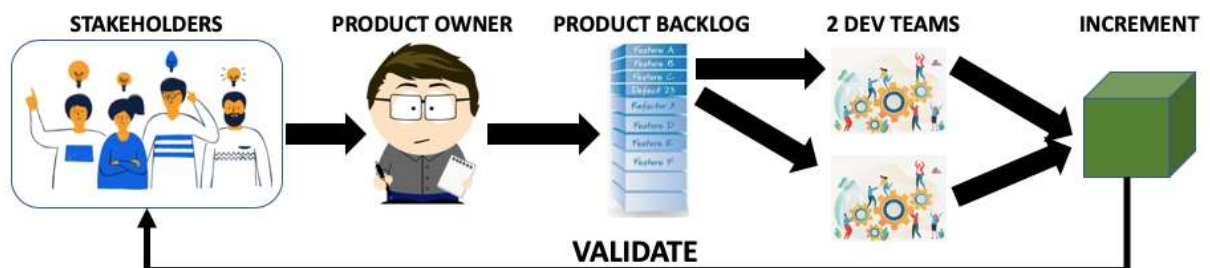
A	• Active Participation in Scrum Events • Available to your Dev team	B	• Be Bold in making right decisions • Be Genuine with Dev team and Stakeholders	C	• Collaborate with Dev Team & Consider yourself as part of the Scrum team • Competitor awareness	D	• Do not push your Development team • Domain expertise • Design thinking	E	• Effective Backlog Management • Engaging • Entrepreneurial	F	• Find ways to optimize the whole • Feedback to teams regularly	G	Good grip on the Product functionality
H	• Honest with Dev team and Stakeholders • "How" is not your part of job	I	• If you cannot Push down, Push up	L	• Long vs Short term trade-offs • Lead the Backlog refinement	M	• Market awareness • MVP oriented	N	"No" word usage	O	Open to receive feedback	P	Passionate about Product Development
Q	Quick decision making	R	Receive regular feedback from stakeholders	S	Scrum knowledge	T	Trust & Transparent with Dev team and stakeholders	U	UX/UI awareness	V	Visionary	W	"We" thinking instead of "I/You"

Scrum Team Setup

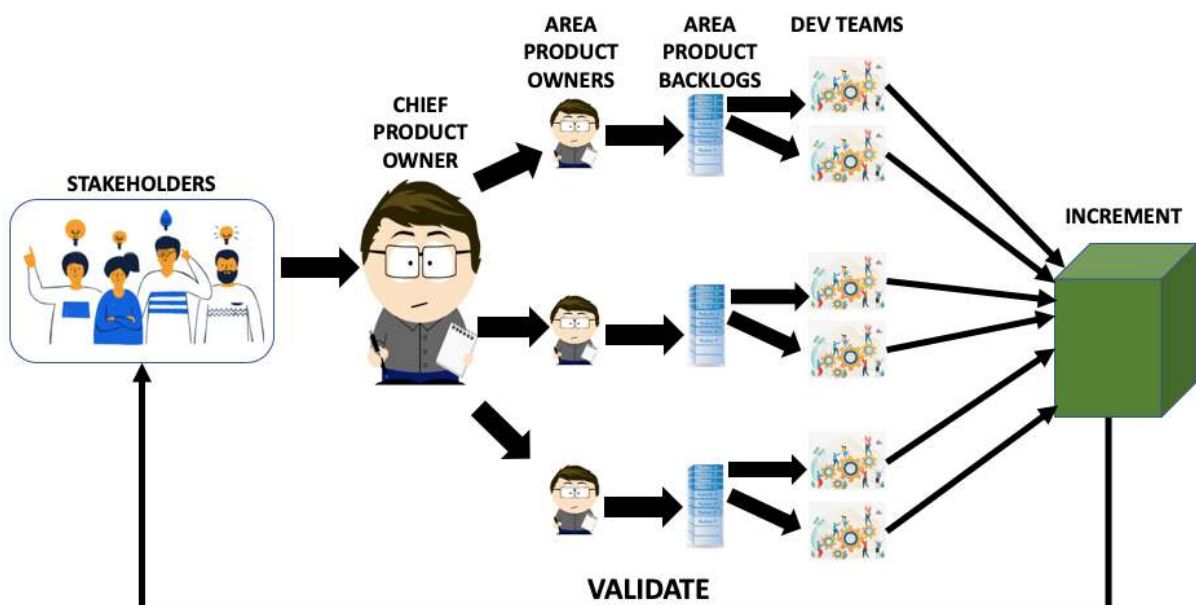
Single Team setup



More than one Team setup



Scaled Team setup

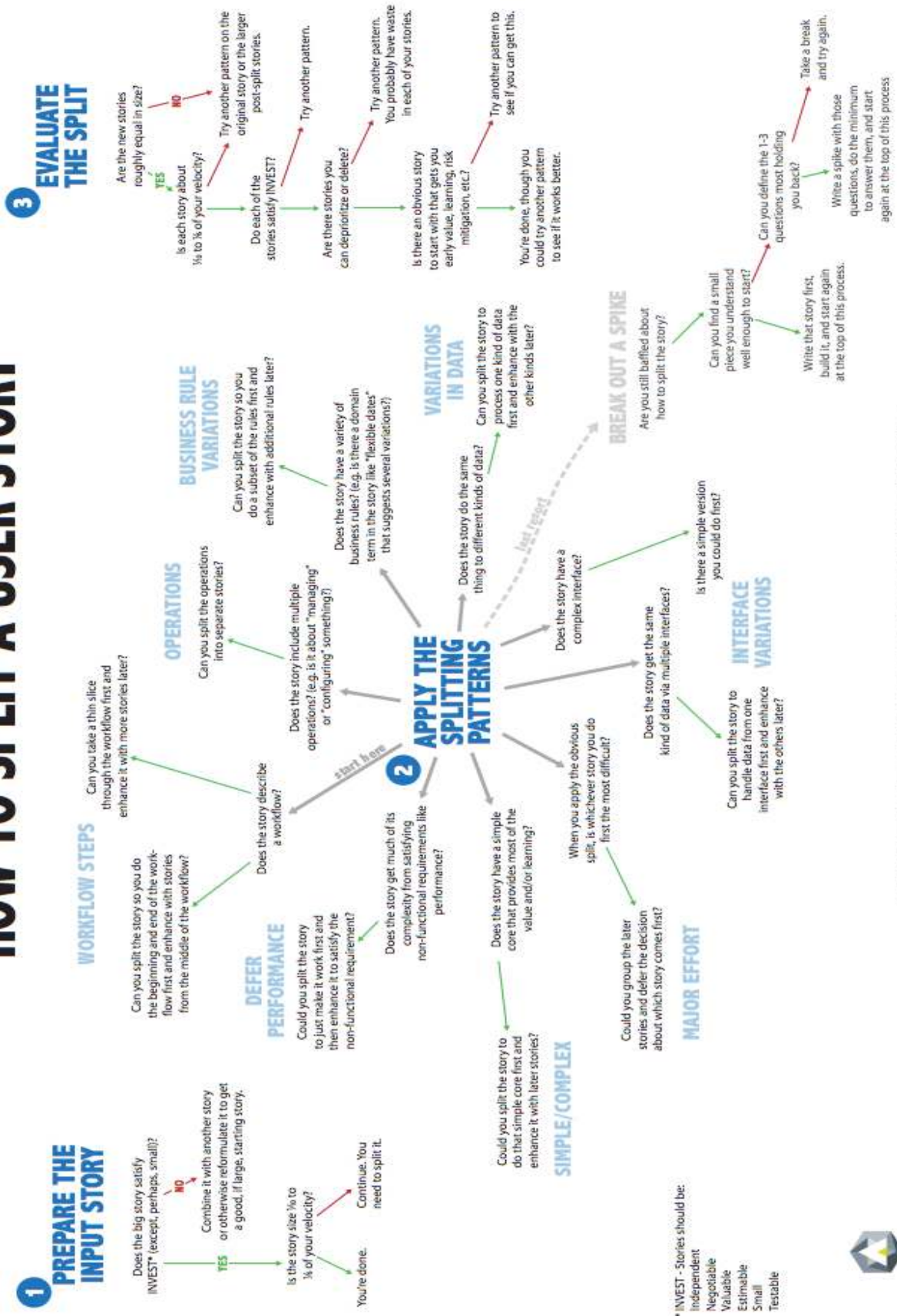


Product Owner Anti-Patterns

1. One PO per Developers instead of one PO per Product
2. Multiple PO for multiple smaller **related** “products” (e.g. PO for Android PowerPoint, PO for iPhone PowerPoint, PO for Windows Desktop PowerPoint, PO for Mac PowerPoint, PO for PowerPoint 365)
3. One PO per feature/component/epic (“feature” product owner, “component” Product Owner, “epic” product owner)
4. One “onsite” PO & one “offshore” PO
5. PO being the bottleneck in communication, acting as in intermediary between the team & stakeholders/customers increasing the number of “hops” in communication
6. PO not having the authority to terminate the Sprint
7. ex-BA now relabelled as PO – nothing changes in the work they do
8. PO not from Product Management or Product Marketing (if building external facing product) or not from business (if building internal products)
9. Relabelling “project managers” to PO
10. “Technical” Product Owner (usually from engineering/IT), “proxy” Product Owner, “functional” Product Owner (i.e. functional managers are ***relabelled*** as Product Owner e.g. “Dev” Product Owner, “QA” Product Owner)
11. PO not being a subject matter/domain expert and act as just order taker
12. PO having multiple Developers level Product Backlog instead of having ONE product backlog and multiple teams pulling from them
13. PO not understanding the customers/users of the Product and not understanding the market
14. Others in the organization not respecting PO’s final decision on the Product Backlog
15. PO spoon-feeding the Developers with Product Backlog items (duh!! seriously?) instead of them writing it
16. PO & Developers member being the same person
17. PO & Scrum Master being the same person
18. PO managing “Projects”, “Programs” etc instead of “Product”
19. No Product Owner assigned to the Product
20. PO not meeting with the Dev. Team face to face periodically (if not co-located)

User Story Splitting Patterns

HOW TO SPLIT A USER STORY



References for further learning

CSPO certification renewal process - <https://www.scrumalliance.org/get-certified/renewing-certifications>

Scrum Guide - <http://www.scrumguides.org/>

Scrum Alliance – <http://www.scrumalliance.org/>

My articles blog – <https://www.learnovative.com/blog/>

My YouTube videos - <https://www.youtube.com/learnovative>

Product Owner Interview Questions and Answers <https://www.learnovative.com/resources>

Linked in user Group – <https://www.linkedin.com/groups/8305379>

Google Drive For Additional Material: <https://tinyurl.com/CSPO-Vijay-Bandaru>



How can you benefited through connecting with us in future?

User Group

We have a Telegram User Group through which you can get connected for various benefits

Resume Review

We will help you to review and suggest improvements to your Resume to make it better

Mock Interview

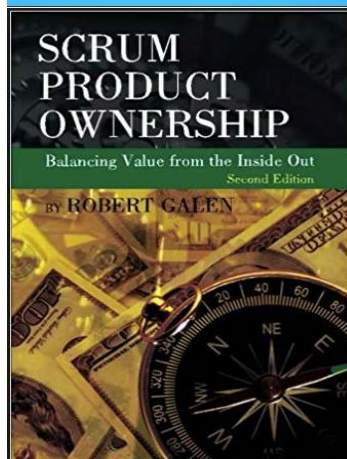
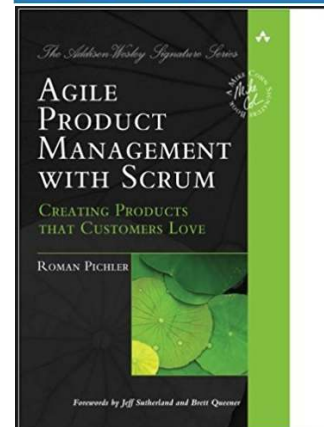
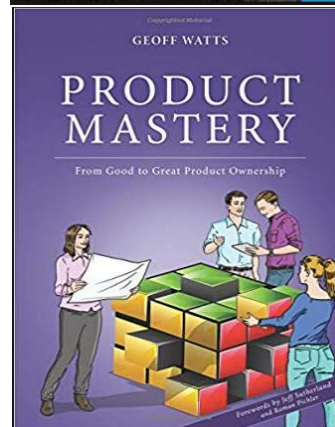
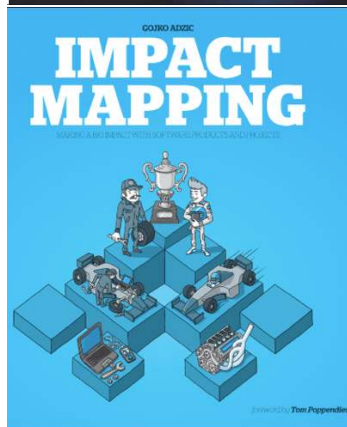
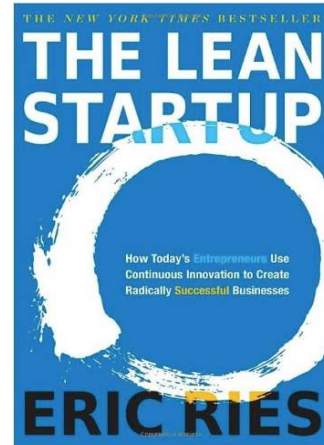
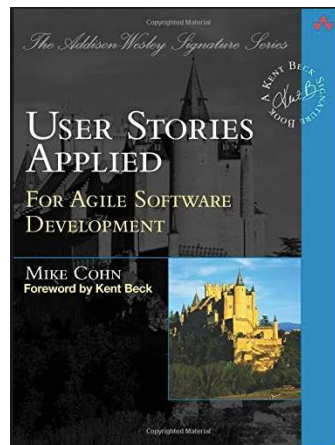
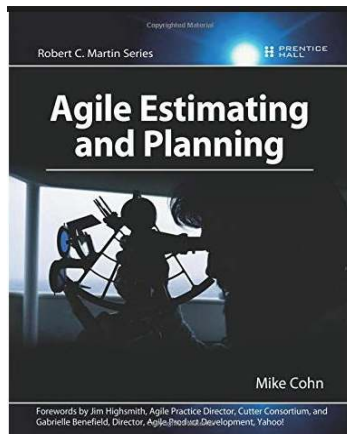
We will help with Mock Interview for your upcoming Product Owner/Product Manager interviews

To avail these benefits, you can reach out to our Team through:

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